

Notebook

July 4, 2025

```
[6]: !pip install scikit-learn
```

```
Requirement already satisfied: scikit-learn in
c:\users\pari\appdata\local\programs\python\python313\lib\site-packages (1.7.0)
Requirement already satisfied: numpy>=1.22.0 in
c:\users\pari\appdata\local\programs\python\python313\lib\site-packages (from
scikit-learn) (2.2.5)
Requirement already satisfied: scipy>=1.8.0 in
c:\users\pari\appdata\local\programs\python\python313\lib\site-packages (from
scikit-learn) (1.16.0)
Requirement already satisfied: joblib>=1.2.0 in
c:\users\pari\appdata\local\programs\python\python313\lib\site-packages (from
scikit-learn) (1.5.1)
Requirement already satisfied: threadpoolctl>=3.1.0 in
c:\users\pari\appdata\local\programs\python\python313\lib\site-packages (from
scikit-learn) (3.6.0)
```

```
[62]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import sklearn
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import confusion_matrix, accuracy_score, \
    classification_report
import warnings
warnings.filterwarnings("ignore")
```

#Basic EDA

```
[4]: df=pd.read_csv('loan_detection.csv')
df
```

```
[4]:
```

	age	campaign	pdays	previous	no_previous_contact	not_working	\
0	56	1	999	0	1	0	
1	57	1	999	0	1	0	
2	37	1	999	0	1	0	

3	40	1	999	0	1	0
4	56	1	999	0	1	0
...	...	...	...	...	...	...
41183	73	1	999	0	1	1
41184	46	1	999	0	1	0
41185	56	2	999	0	1	1
41186	44	1	999	0	1	0
41187	74	3	999	1	1	1

	job_admin.	job_blue-collar	job_entrepreneur	job_housemaid	...	\
0	0	0	0	1	...	
1	0	0	0	0	...	
2	0	0	0	0	...	
3	1	0	0	0	...	
4	0	0	0	0	...	
...	...	...	...	...	...	
41183	0	0	0	0	...	
41184	0	1	0	0	...	
41185	0	0	0	0	...	
41186	0	0	0	0	...	
41187	0	0	0	0	...	

	month_sep	day_of_week_fri	day_of_week_mon	day_of_week_thu	\
0	0	0	1	0	
1	0	0	1	0	
2	0	0	1	0	
3	0	0	1	0	
4	0	0	1	0	
...	...	...	...	...	
41183	0	1	0	0	
41184	0	1	0	0	
41185	0	1	0	0	
41186	0	1	0	0	
41187	0	1	0	0	

	day_of_week_tue	day_of_week_wed	poutcome_failure	\
0	0	0	0	
1	0	0	0	
2	0	0	0	
3	0	0	0	
4	0	0	0	
...	...	...	...	
41183	0	0	0	
41184	0	0	0	
41185	0	0	0	
41186	0	0	0	
41187	0	0	1	

	poutcome_nonexistent	poutcome_success	Loan_Status_label
0	1	0	0
1	1	0	0
2	1	0	0
3	1	0	0
4	1	0	0
...	...	...	...
41183	1	0	1
41184	1	0	0
41185	1	0	0
41186	1	0	1
41187	0	0	0

[41188 rows x 60 columns]

#Statistical Analysis

```
[10]: df.shape
```

```
[10]: (41188, 60)
```

```
[11]: df.columns
```

```
[11]: Index(['age', 'campaign', 'pdays', 'previous', 'no_previous_contact',
            'not_working', 'job_admin.', 'job_blue-collar', 'job_entrepreneur',
            'job_housemaid', 'job_management', 'job_retired', 'job_self-employed',
            'job_services', 'job_student', 'job_technician', 'job_unemployed',
            'job_unknown', 'marital_divorced', 'marital_married', 'marital_single',
            'marital_unknown', 'education_basic.4y', 'education_basic.6y',
            'education_basic.9y', 'education_high.school', 'education_illiterate',
            'education_professional.course', 'education_university.degree',
            'education_unknown', 'default_no', 'default_unknown', 'default_yes',
            'housing_no', 'housing_unknown', 'housing_yes', 'loan_no',
            'loan_unknown', 'loan_yes', 'contact_cellular', 'contact_telephone',
            'month_apr', 'month_aug', 'month_dec', 'month_jul', 'month_jun',
            'month_mar', 'month_may', 'month_nov', 'month_oct', 'month_sep',
            'day_of_week_fri', 'day_of_week_mon', 'day_of_week_thu',
            'day_of_week_tue', 'day_of_week_wed', 'poutcome_failure',
            'poutcome_nonexistent', 'poutcome_success', 'Loan_Status_label'],
            dtype='object')
```

```
[12]: df.info
```

```
[12]: <bound method DataFrame.info of          age  campaign  pdays  previous
no_previous_contact  not_working  \
0          56          1    999          0          1          0
1          57          1    999          0          1          0
```

2	37	1	999	0	1	0
3	40	1	999	0	1	0
4	56	1	999	0	1	0
...	...	...	...	...	...	...
41183	73	1	999	0	1	1
41184	46	1	999	0	1	0
41185	56	2	999	0	1	1
41186	44	1	999	0	1	0
41187	74	3	999	1	1	1

	job_admin.	job_blue-collar	job_entrepreneur	job_housemaid	...	\
0	0	0	0	1	...	
1	0	0	0	0	...	
2	0	0	0	0	...	
3	1	0	0	0	...	
4	0	0	0	0	...	
...	...	...	...	...	...	
41183	0	0	0	0	...	
41184	0	1	0	0	...	
41185	0	0	0	0	...	
41186	0	0	0	0	...	
41187	0	0	0	0	...	

	month_sep	day_of_week_fri	day_of_week_mon	day_of_week_thu	\
0	0	0	1	0	
1	0	0	1	0	
2	0	0	1	0	
3	0	0	1	0	
4	0	0	1	0	
...	...	...	...	...	
41183	0	1	0	0	
41184	0	1	0	0	
41185	0	1	0	0	
41186	0	1	0	0	
41187	0	1	0	0	

	day_of_week_tue	day_of_week_wed	poutcome_failure	\
0	0	0	0	
1	0	0	0	
2	0	0	0	
3	0	0	0	
4	0	0	0	
...	...	...	...	
41183	0	0	0	
41184	0	0	0	
41185	0	0	0	
41186	0	0	0	

	0	0	1
	poutcome_nonexistent	poutcome_success	Loan_Status_label
0	1	0	0
1	1	0	0
2	1	0	0
3	1	0	0
4	1	0	0
...	...	...	...
41183	1	0	1
41184	1	0	0
41185	1	0	0
41186	1	0	1
41187	0	0	0

[41188 rows x 60 columns]>

```
[13]: df.describe()
```

```
[13]:
```

	age	campaign	pdays	previous \
count	41188.000000	41188.000000	41188.000000	41188.000000
mean	40.02406	2.567593	962.475454	0.172963
std	10.42125	2.770014	186.910907	0.494901
min	17.000000	1.000000	0.000000	0.000000
25%	32.000000	1.000000	999.000000	0.000000
50%	38.000000	2.000000	999.000000	0.000000
75%	47.000000	3.000000	999.000000	0.000000
max	98.000000	56.000000	999.000000	7.000000

	no_previous_contact	not_working	job_admin.	job_blue-collar \
count	41188.000000	41188.000000	41188.000000	41188.000000
mean	0.963217	0.087623	0.253035	0.224677
std	0.188230	0.282749	0.434756	0.417375
min	0.000000	0.000000	0.000000	0.000000
25%	1.000000	0.000000	0.000000	0.000000
50%	1.000000	0.000000	0.000000	0.000000
75%	1.000000	0.000000	1.000000	0.000000
max	1.000000	1.000000	1.000000	1.000000

	job_entrepreneur	job_housemaid	...	month_sep	day_of_week_fri \
count	41188.000000	41188.000000	...	41188.000000	41188.000000
mean	0.035350	0.025736	...	0.013839	0.190031
std	0.184665	0.158348	...	0.116824	0.392330
min	0.000000	0.000000	...	0.000000	0.000000
25%	0.000000	0.000000	...	0.000000	0.000000
50%	0.000000	0.000000	...	0.000000	0.000000
75%	0.000000	0.000000	...	0.000000	0.000000

max	1.000000	1.000000	...	1.000000	1.000000
-----	----------	----------	-----	----------	----------

	day_of_week_mon	day_of_week_thu	day_of_week_tue	day_of_week_wed	\
count	41188.000000	41188.000000	41188.000000	41188.000000	
mean	0.206711	0.209357	0.196416	0.197485	
std	0.404951	0.406855	0.397292	0.398106	
min	0.000000	0.000000	0.000000	0.000000	
25%	0.000000	0.000000	0.000000	0.000000	
50%	0.000000	0.000000	0.000000	0.000000	
75%	0.000000	0.000000	0.000000	0.000000	
max	1.000000	1.000000	1.000000	1.000000	

	poutcome_failure	poutcome_nonexistent	poutcome_success	\
count	41188.000000	41188.000000	41188.000000	
mean	0.103234	0.863431	0.033335	
std	0.304268	0.343396	0.179512	
min	0.000000	0.000000	0.000000	
25%	0.000000	1.000000	0.000000	
50%	0.000000	1.000000	0.000000	
75%	0.000000	1.000000	0.000000	
max	1.000000	1.000000	1.000000	

	Loan_Status_label
count	41188.000000
mean	0.112654
std	0.316173
min	0.000000
25%	0.000000
50%	0.000000
75%	0.000000
max	1.000000

[8 rows x 60 columns]

Missing values

```
[14]: df.isnull().sum()
```

```
[14]: age                                0
      campaign                           0
      pdays                             0
      previous                           0
      no_previous_contact                 0
      not_working                         0
      job_admin.                         0
      job_blue-collar                    0
      job_entrepreneur                    0
      job_housemaid                       0
```

job_management	0
job_retired	0
job_self-employed	0
job_services	0
job_student	0
job_technician	0
job_unemployed	0
job_unknown	0
marital_divorced	0
marital_married	0
marital_single	0
marital_unknown	0
education_basic.4y	0
education_basic.6y	0
education_basic.9y	0
education_high.school	0
education_illiterate	0
education_professional.course	0
education_university.degree	0
education_unknown	0
default_no	0
default_unknown	0
default_yes	0
housing_no	0
housing_unknown	0
housing_yes	0
loan_no	0
loan_unknown	0
loan_yes	0
contact_cellular	0
contact_telephone	0
month_apr	0
month_aug	0
month_dec	0
month_jul	0
month_jun	0
month_mar	0
month_may	0
month_nov	0
month_oct	0
month_sep	0
day_of_week_fri	0
day_of_week_mon	0
day_of_week_thu	0
day_of_week_tue	0
day_of_week_wed	0
poutcome_failure	0

```
poutcome_nonexistent      0
poutcome_success           0
Loan_Status_label          0
dtype: int64
```

```
[ ]: #No missing values
```

Duplicate values

```
[6]: df.duplicated().sum()
```

```
[6]: np.int64(2417)
```

```
[7]: df.drop_duplicates(keep='first',inplace=True)
```

```
[8]: df.duplicated().sum()
```

```
[8]: np.int64(0)
```

```
[ ]:
```

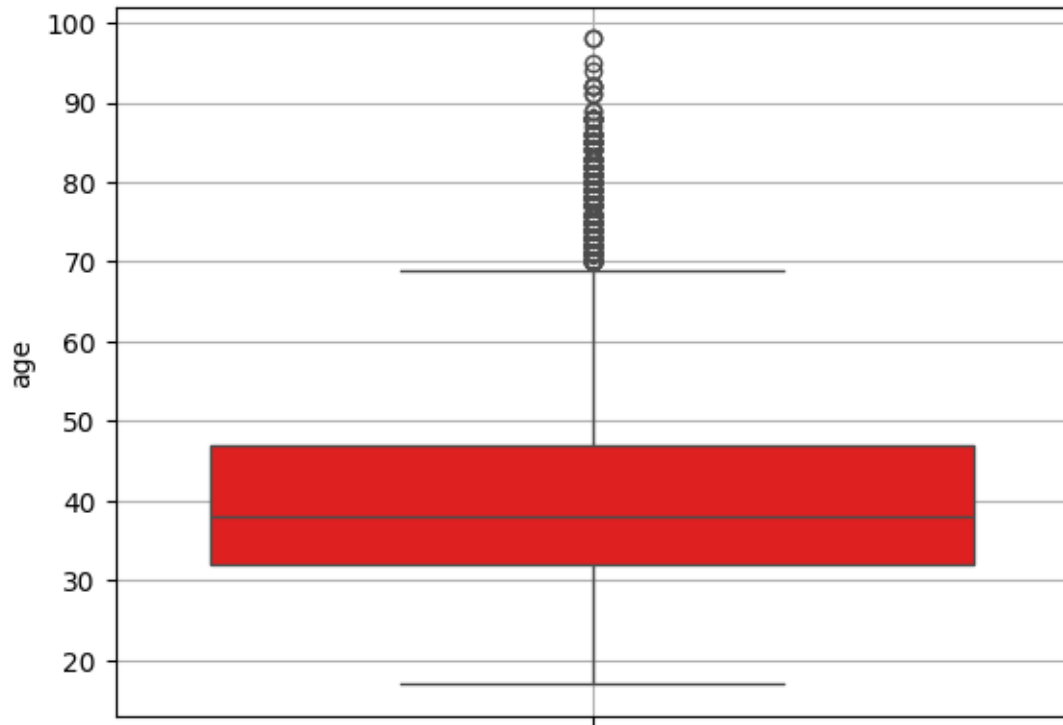
#Outliers

```
[31]: #boxplot
plt.figure(figsize=(15, 15))
sns.boxplot(df)
plt.xticks(rotation=90)
plt.yticks(np.arange(0, 600, step=100))
plt.show()
```

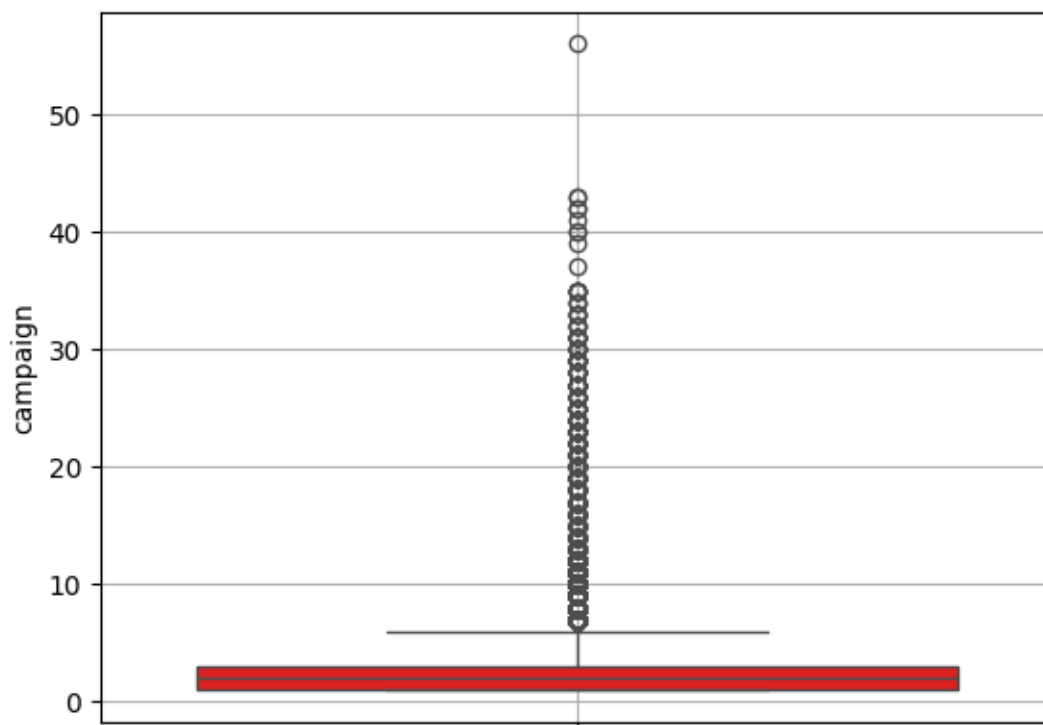


```
plt.grid()  
plt.show()
```

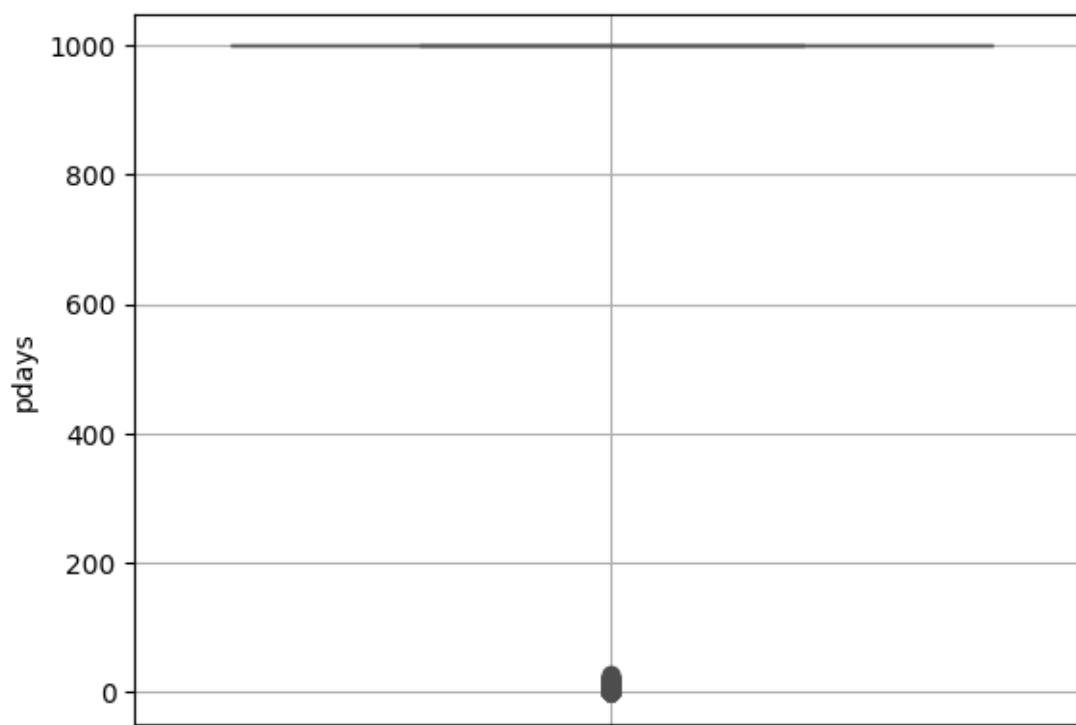
----- age -----



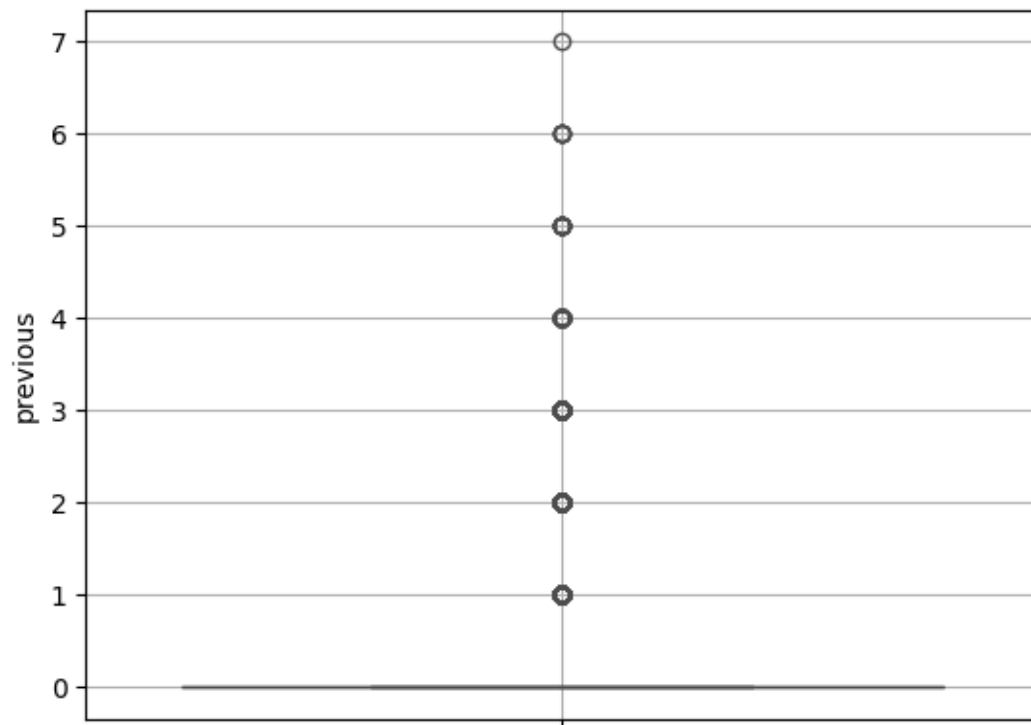
----- campaign -----



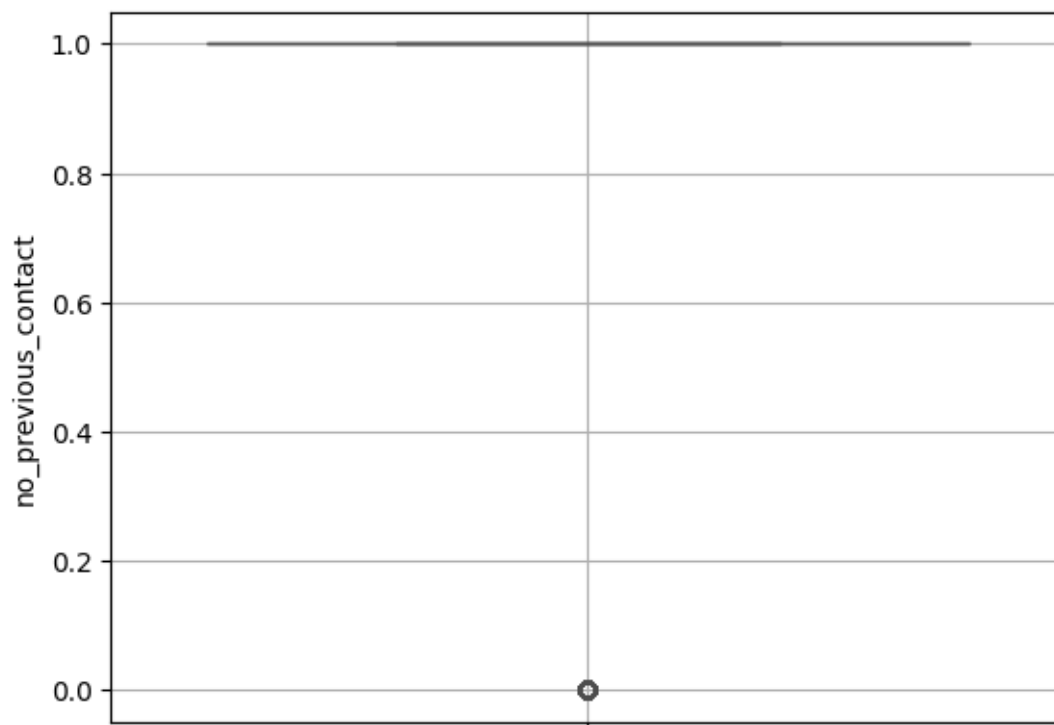
----- pdays -----



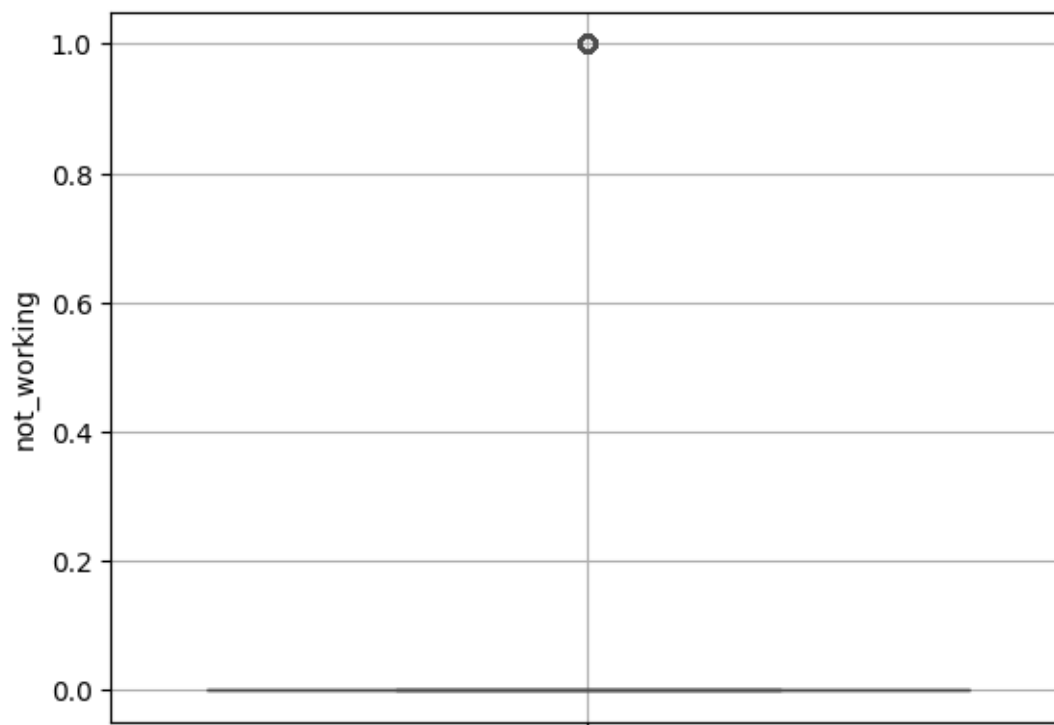
----- previous -----



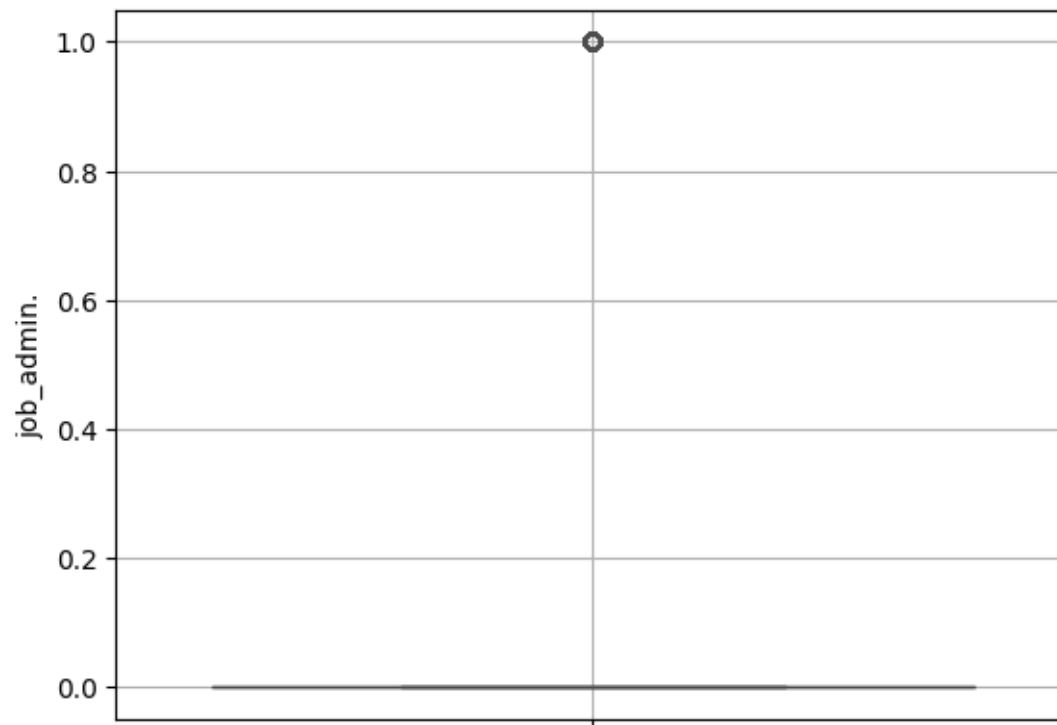
----- no_previous_contact -----



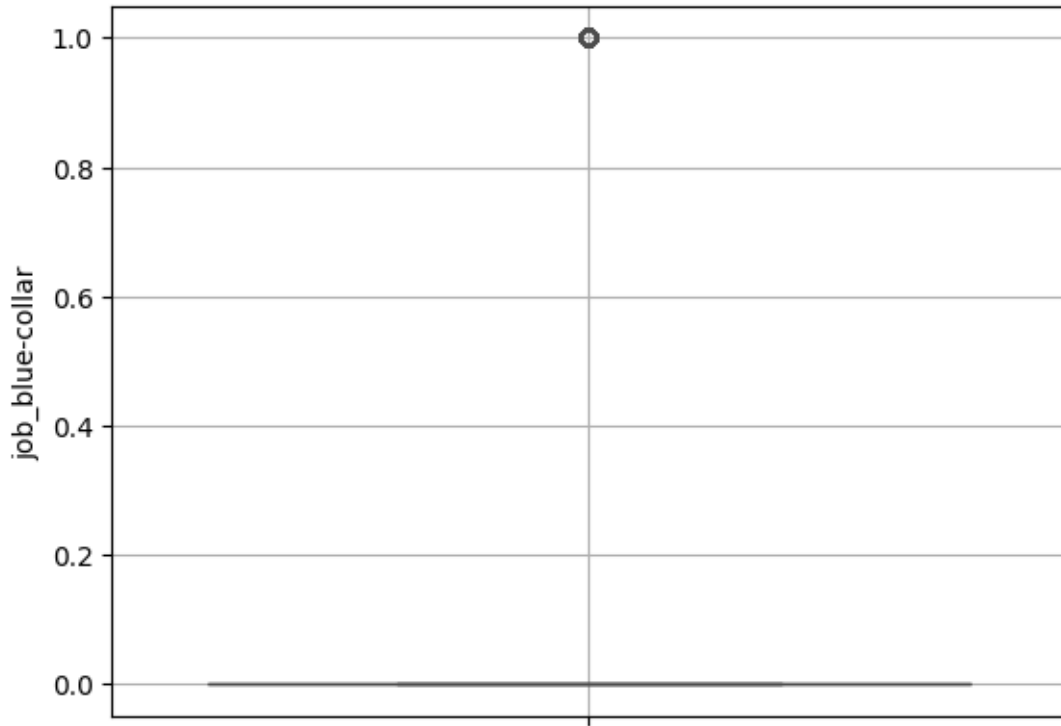
----- not_working -----



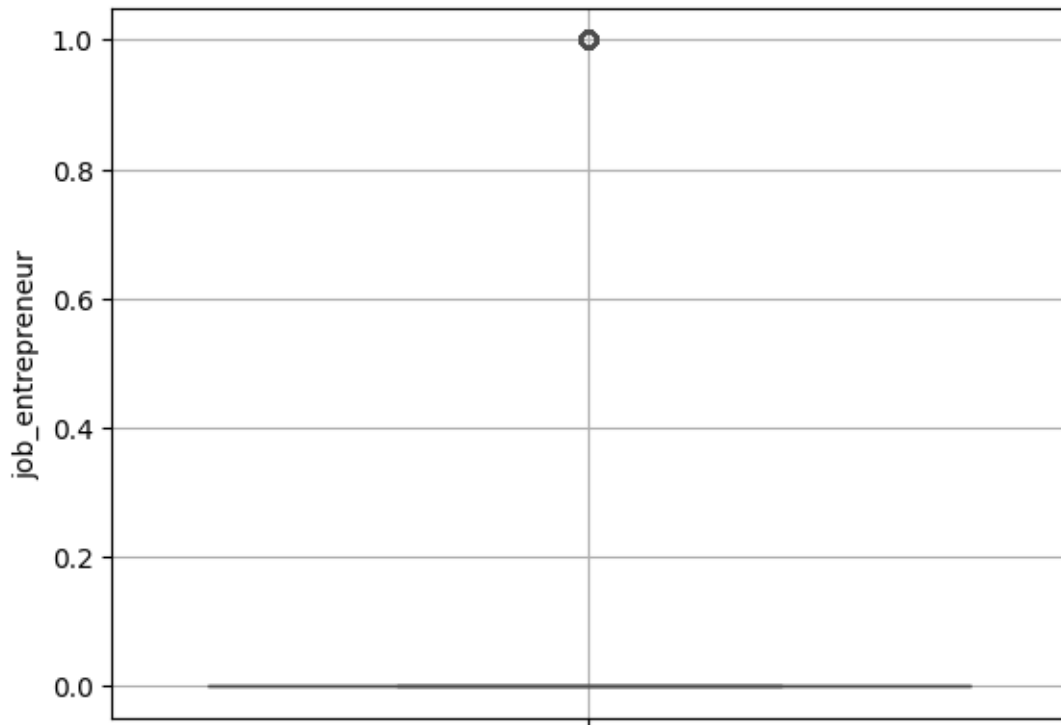
----- job_admin. -----



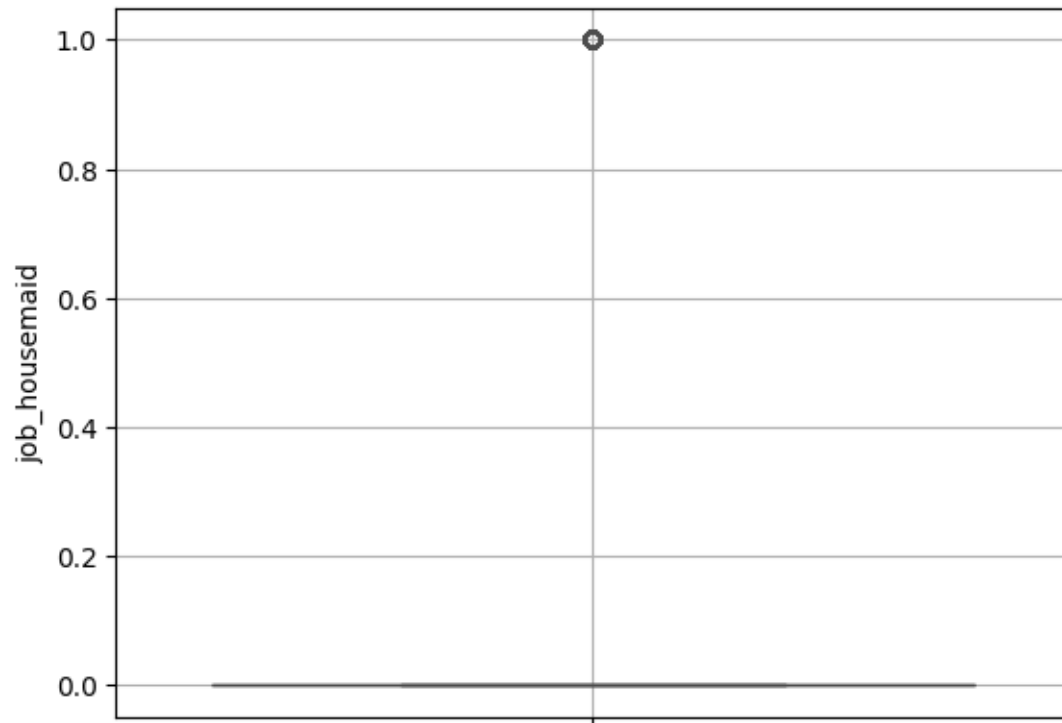
----- job_blue-collar -----



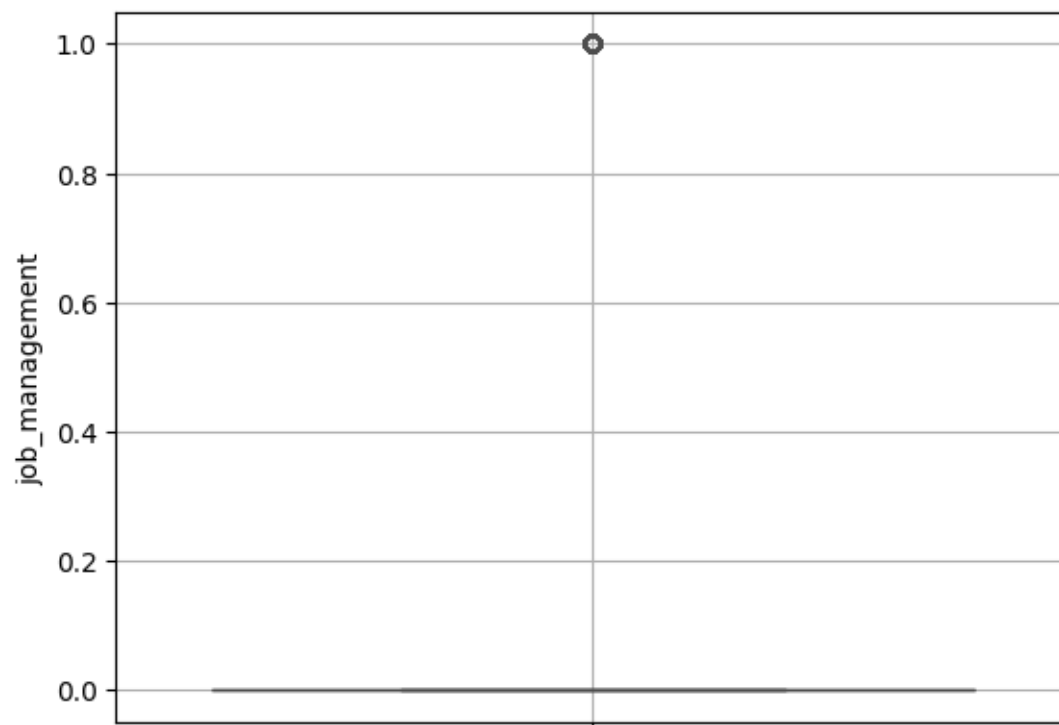
----- job_entrepreneur -----



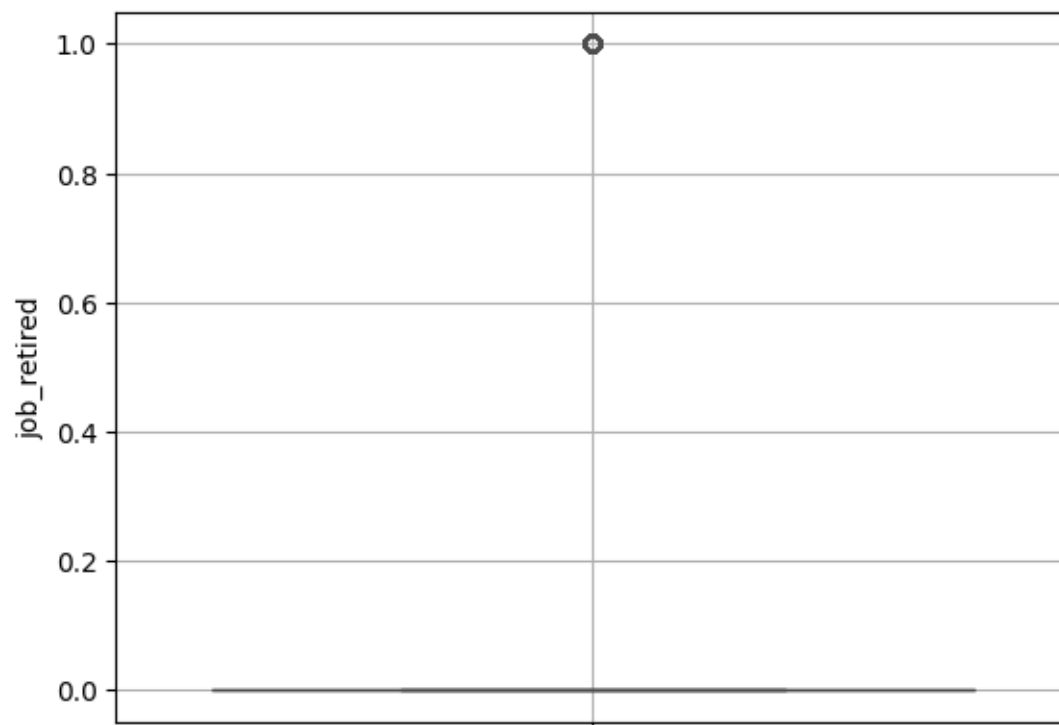
----- job_housemaid -----



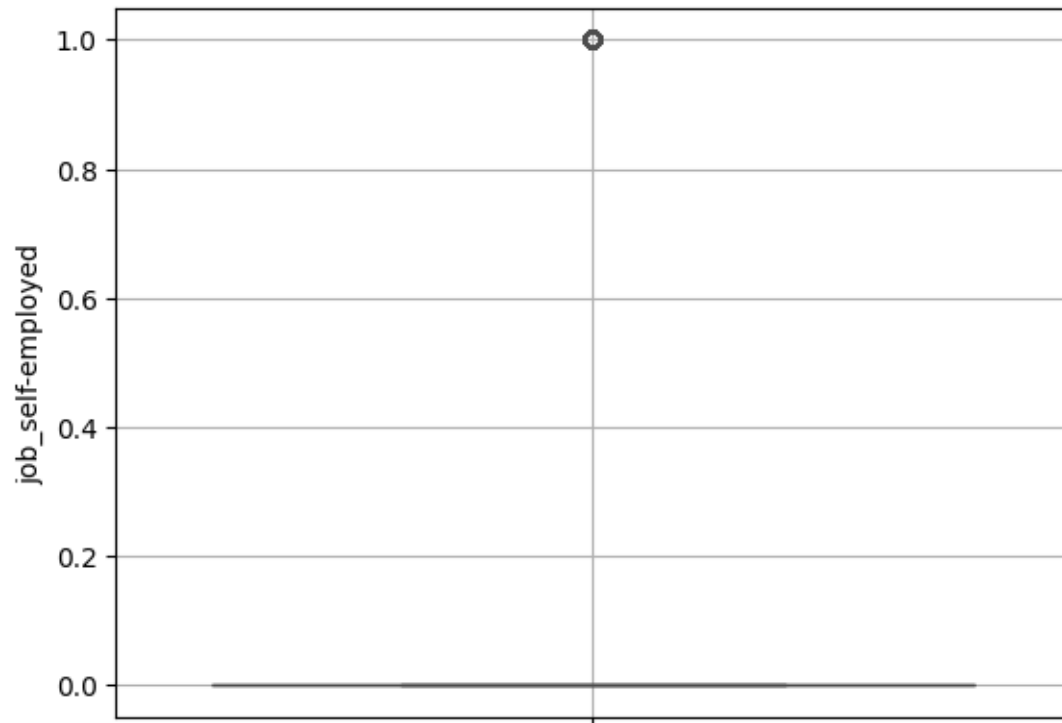
----- job_management -----



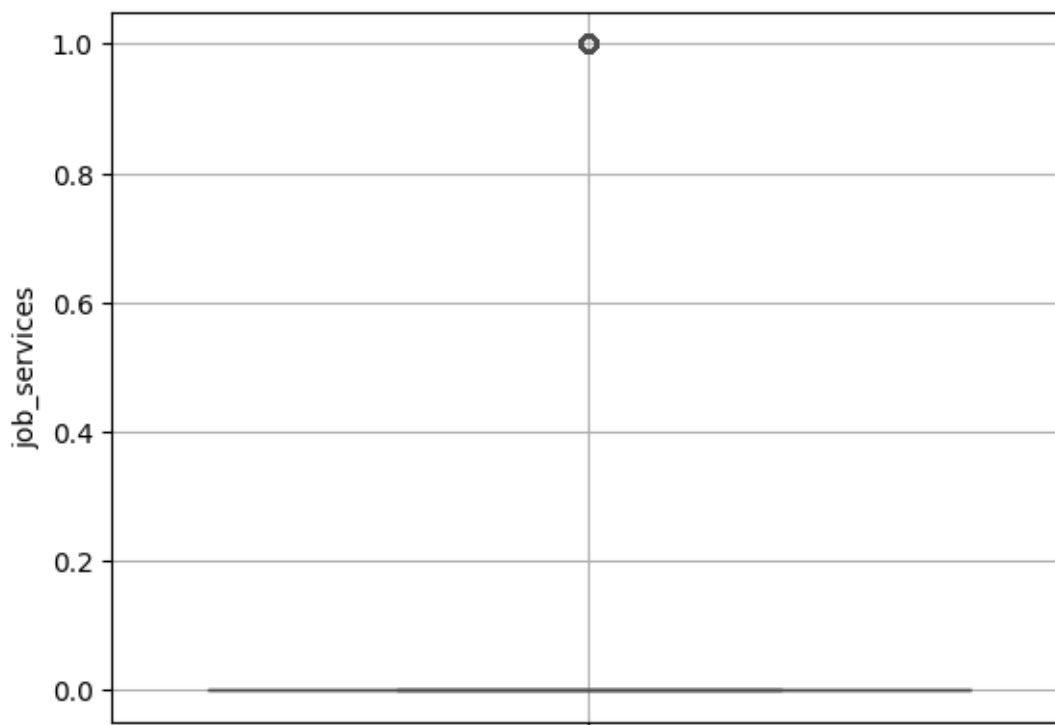
----- job_retired -----



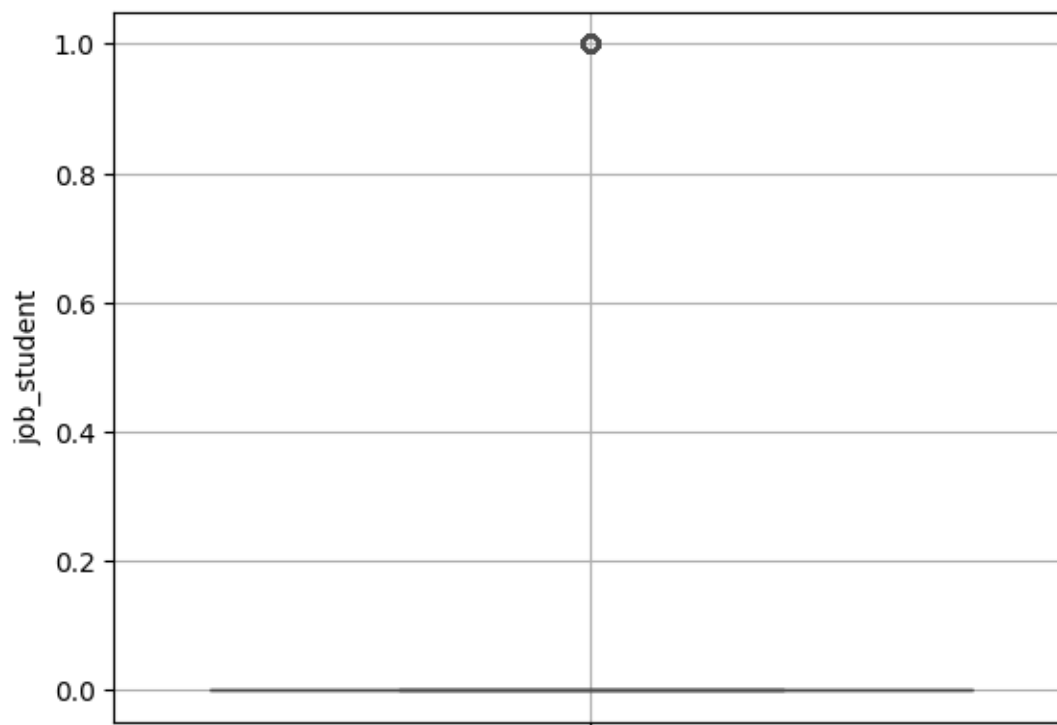
----- job_self-employed -----



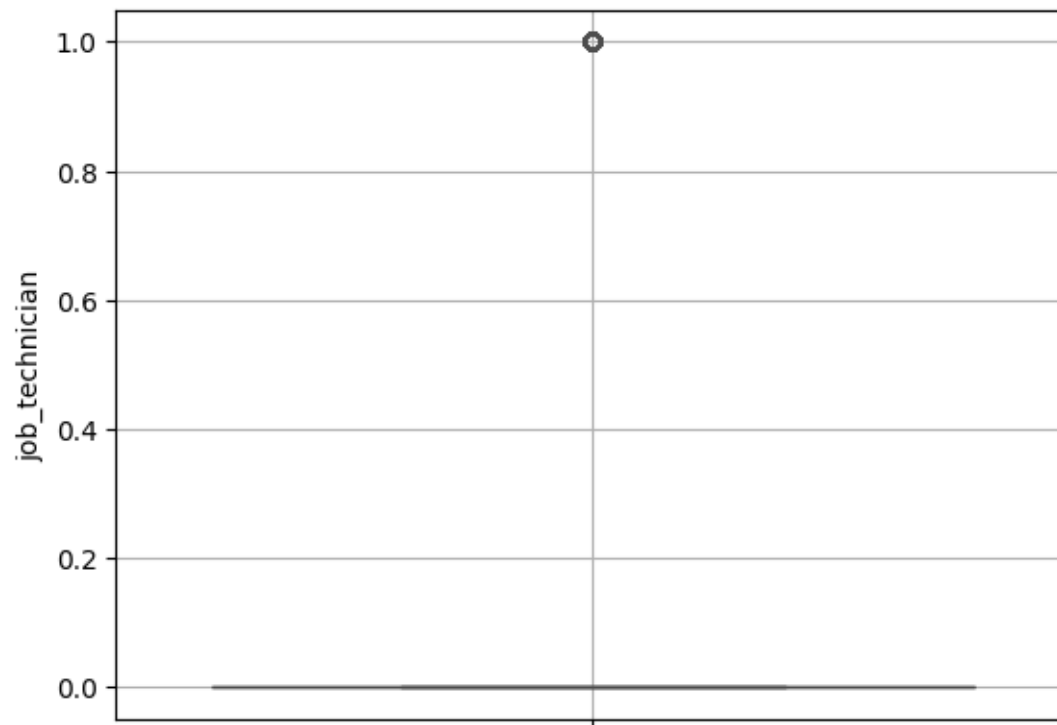
----- job_services -----



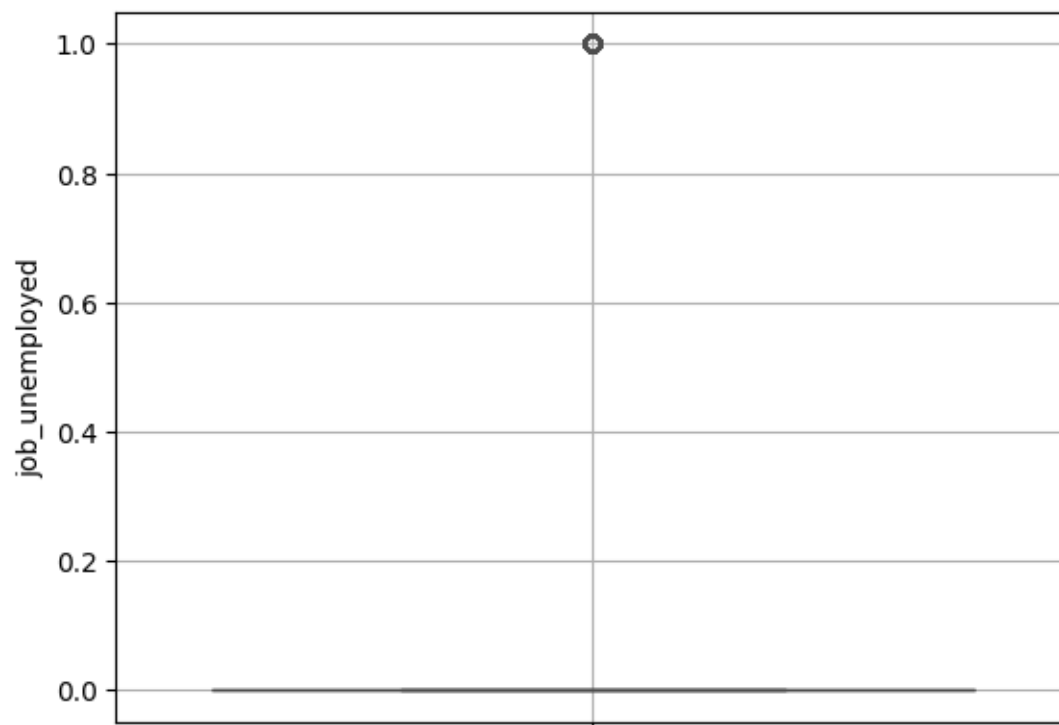
----- job_student -----



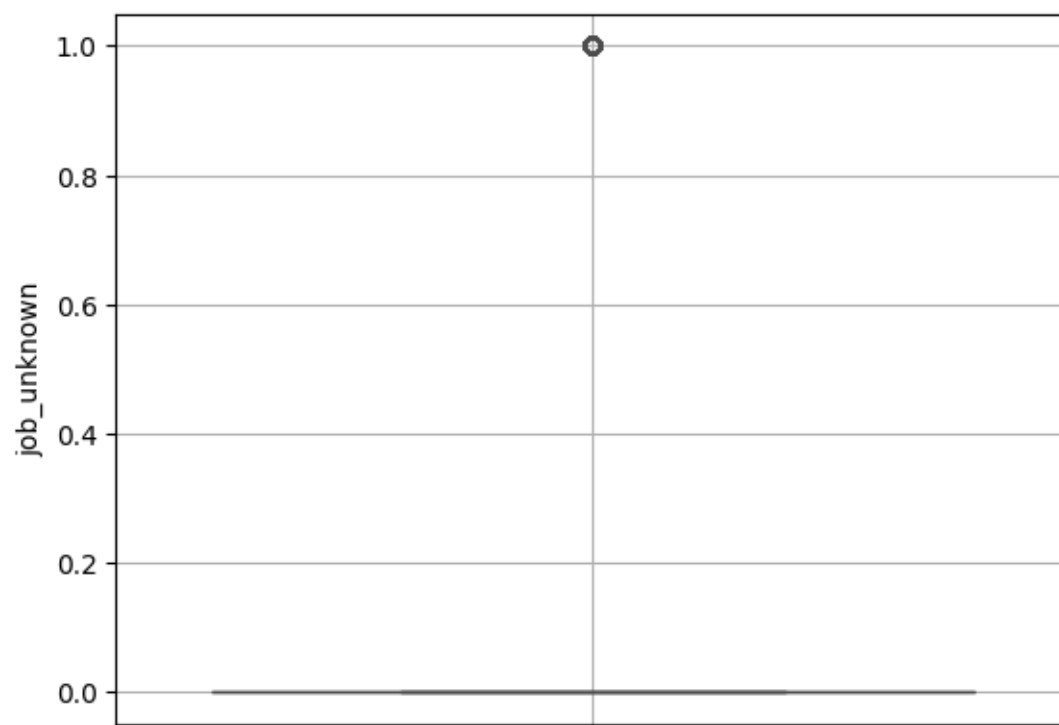
----- job_technician -----



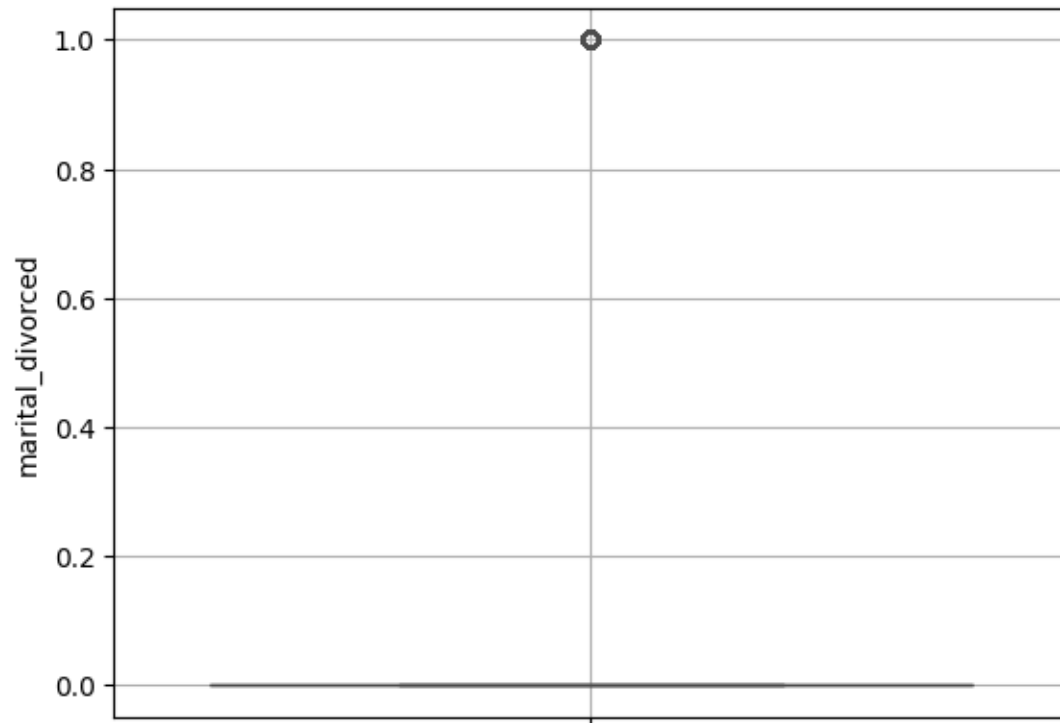
----- job_unemployed -----



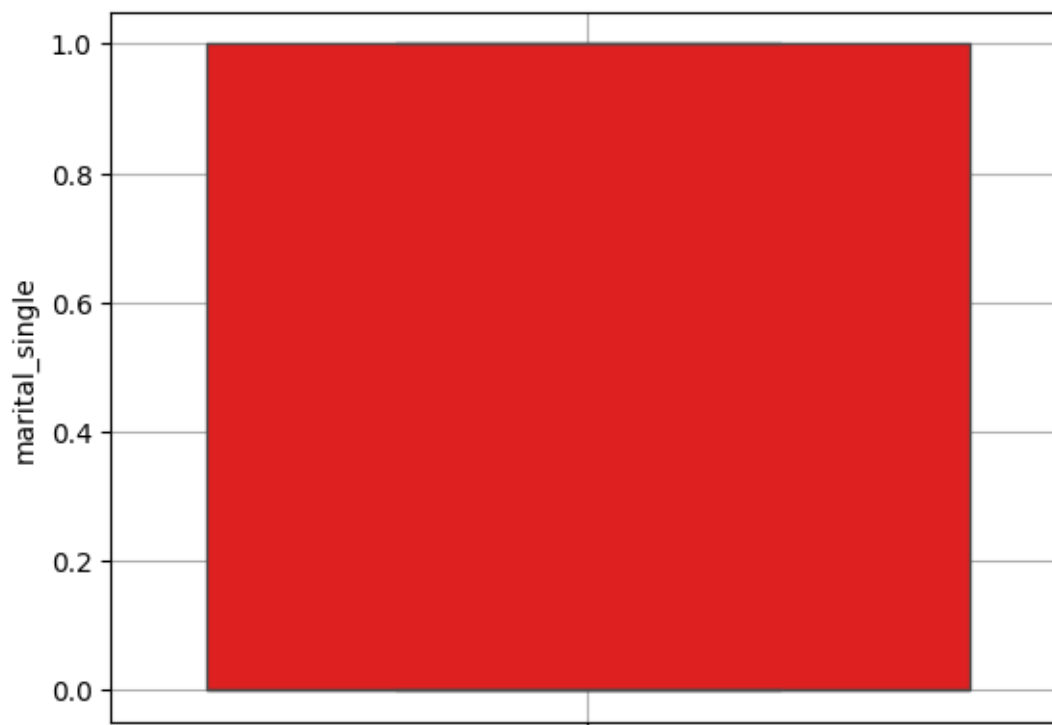
----- job_unknown -----



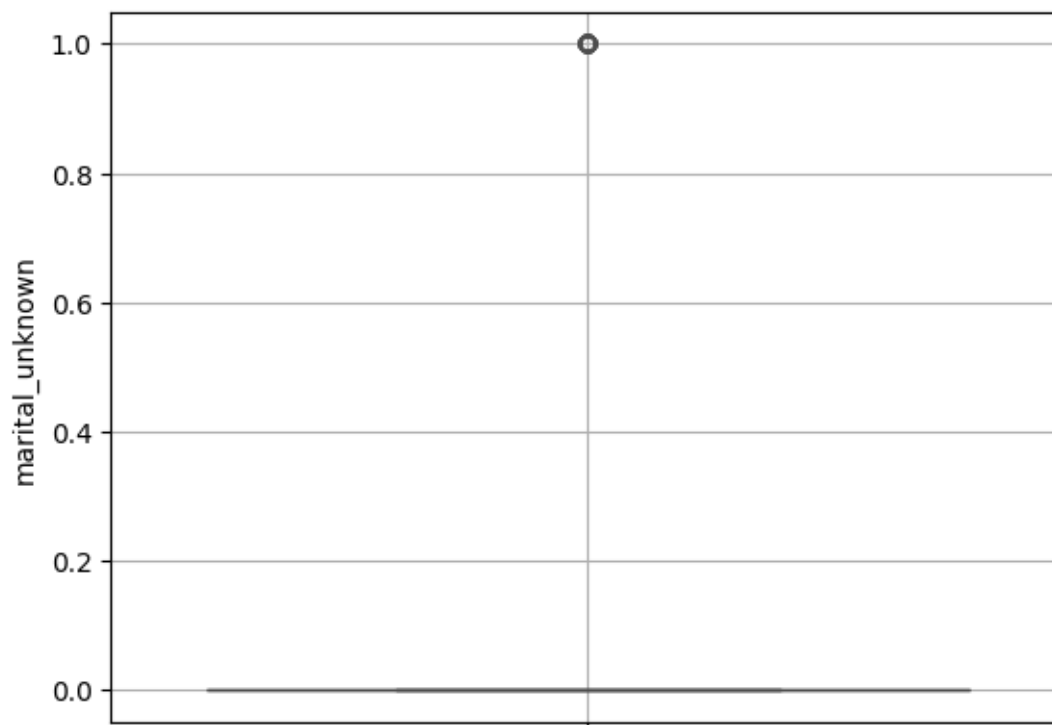
----- marital_divorced -----



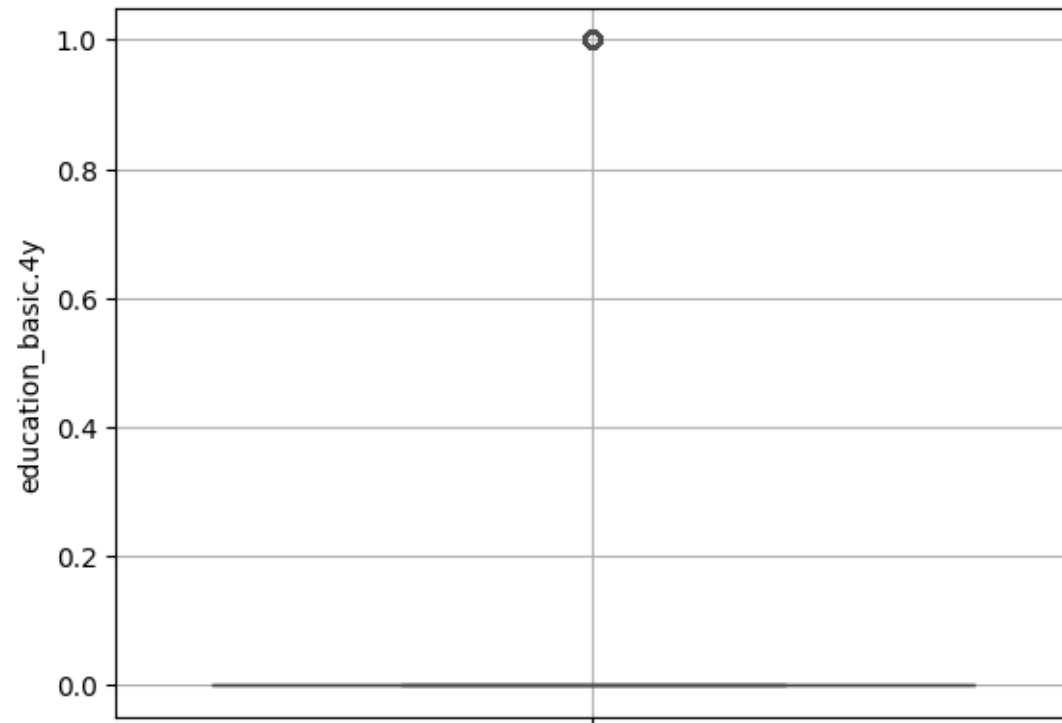
----- marital_single -----



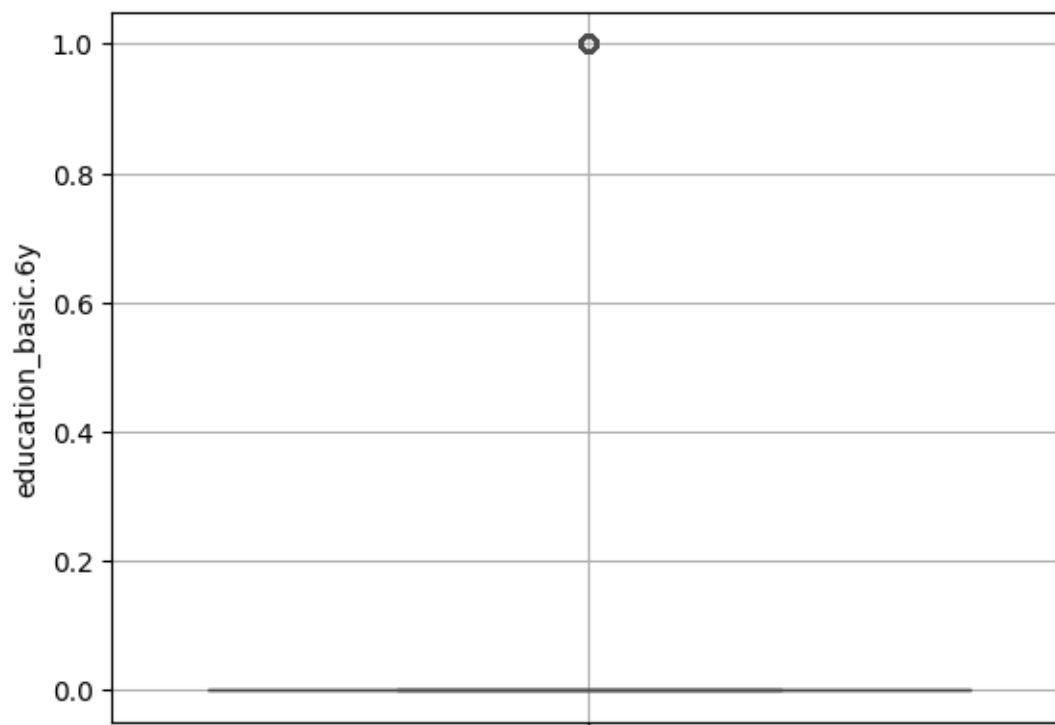
----- marital_unknown -----



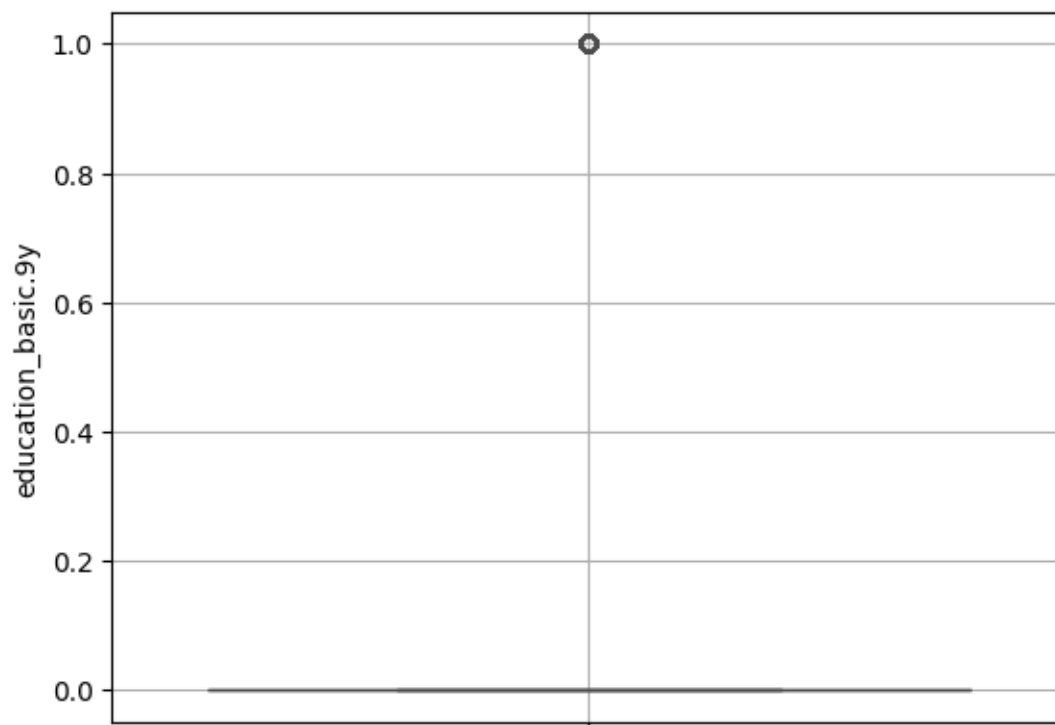
----- education_basic.4y -----



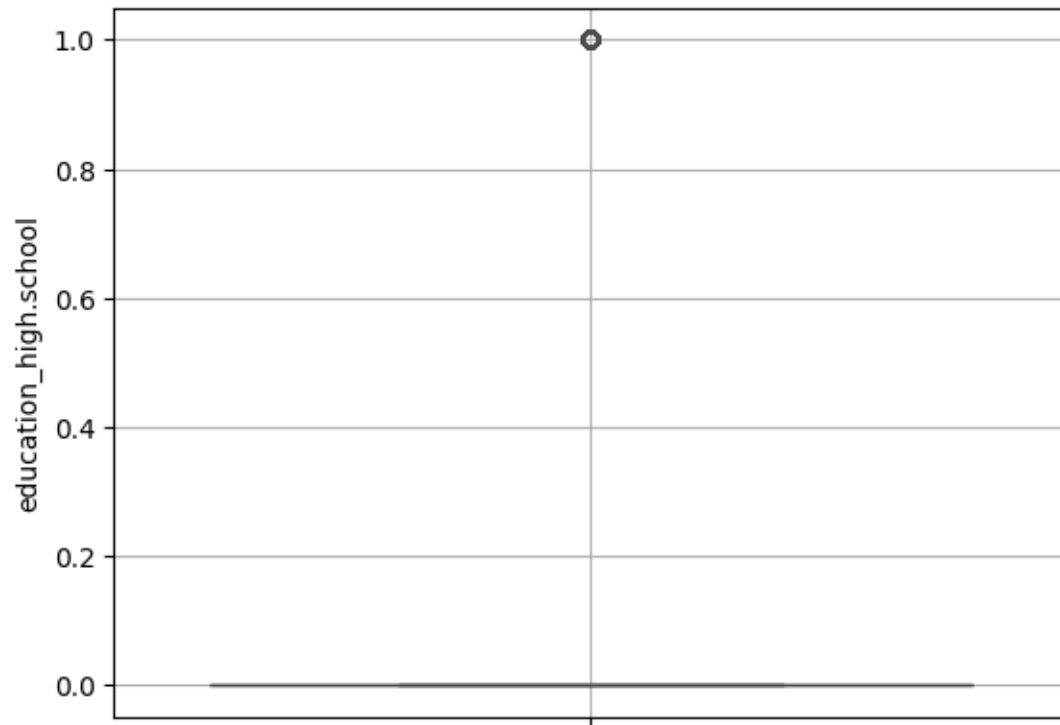
----- education_basic.6y -----



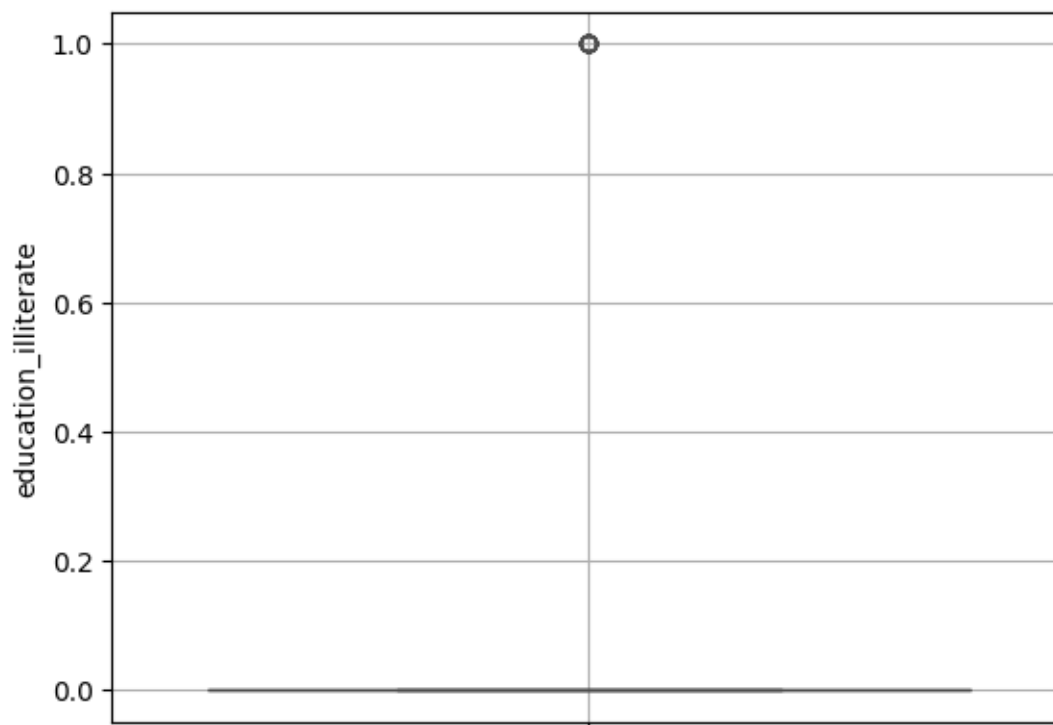
----- education_basic.9y -----



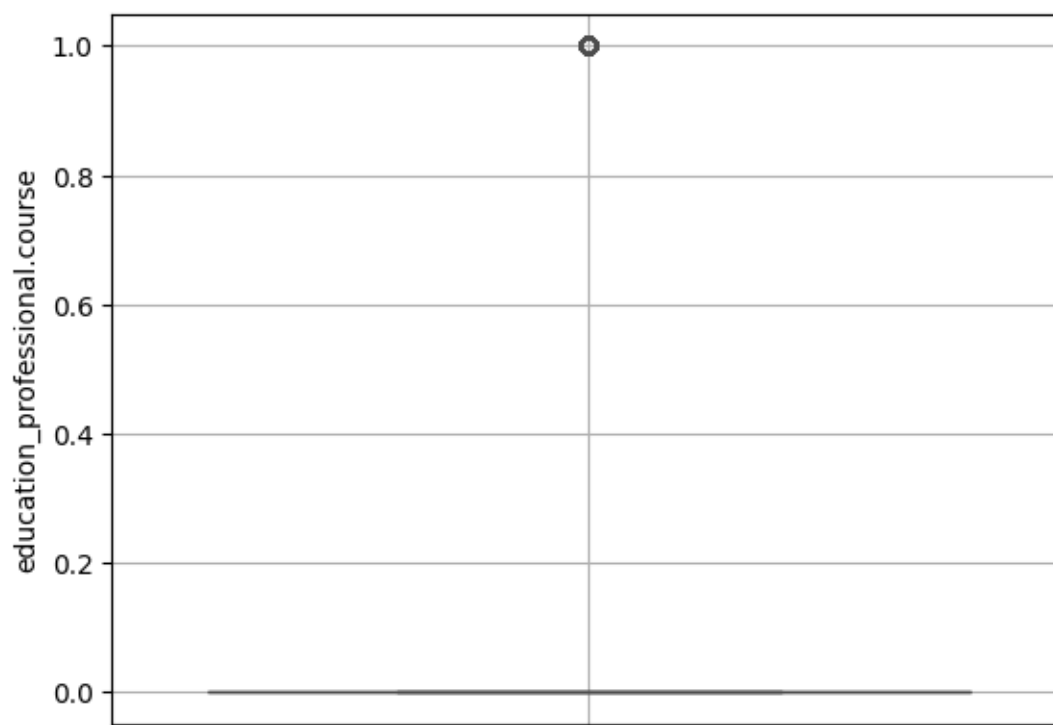
----- education_high.school -----



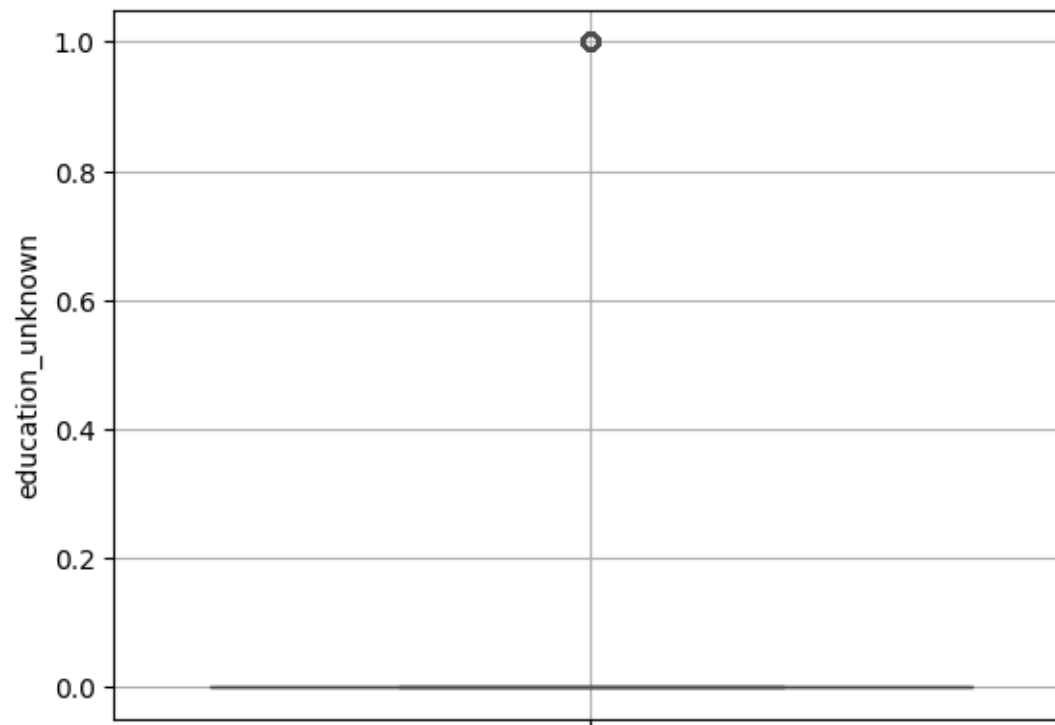
----- education_illiterate -----



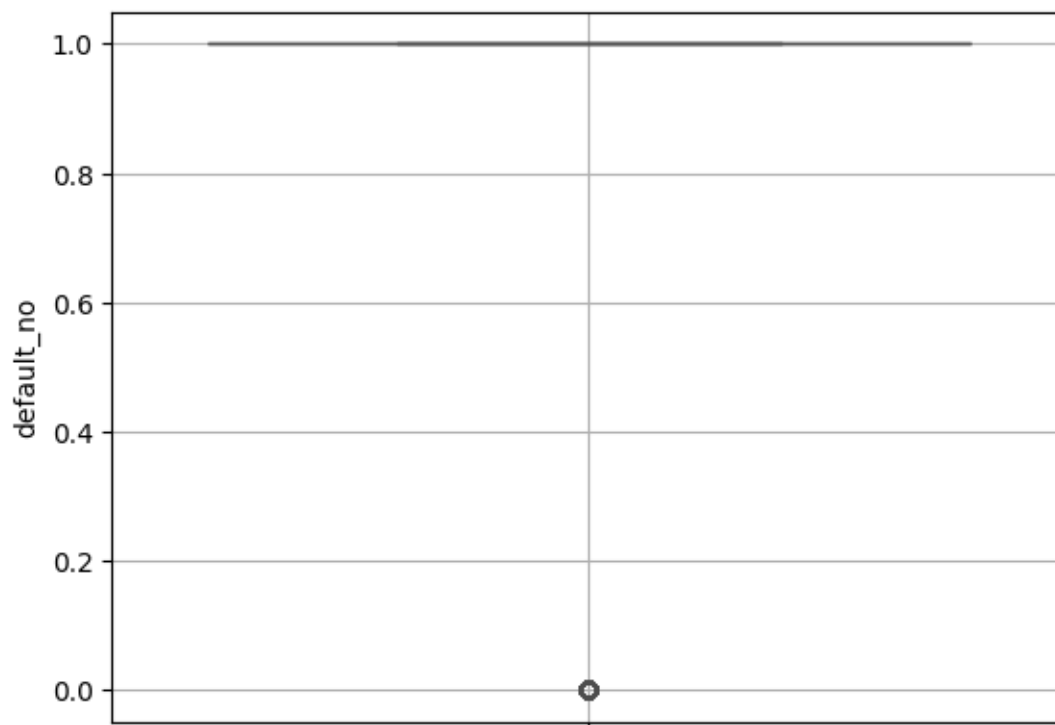
----- education_professional.course -----



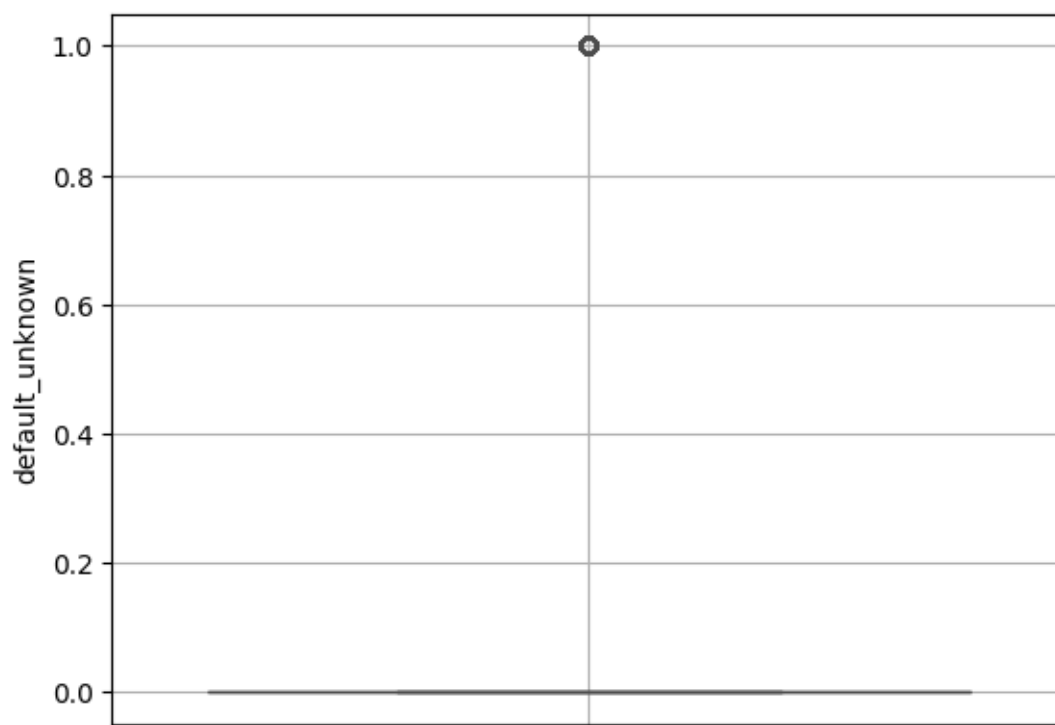
----- education_unknown -----



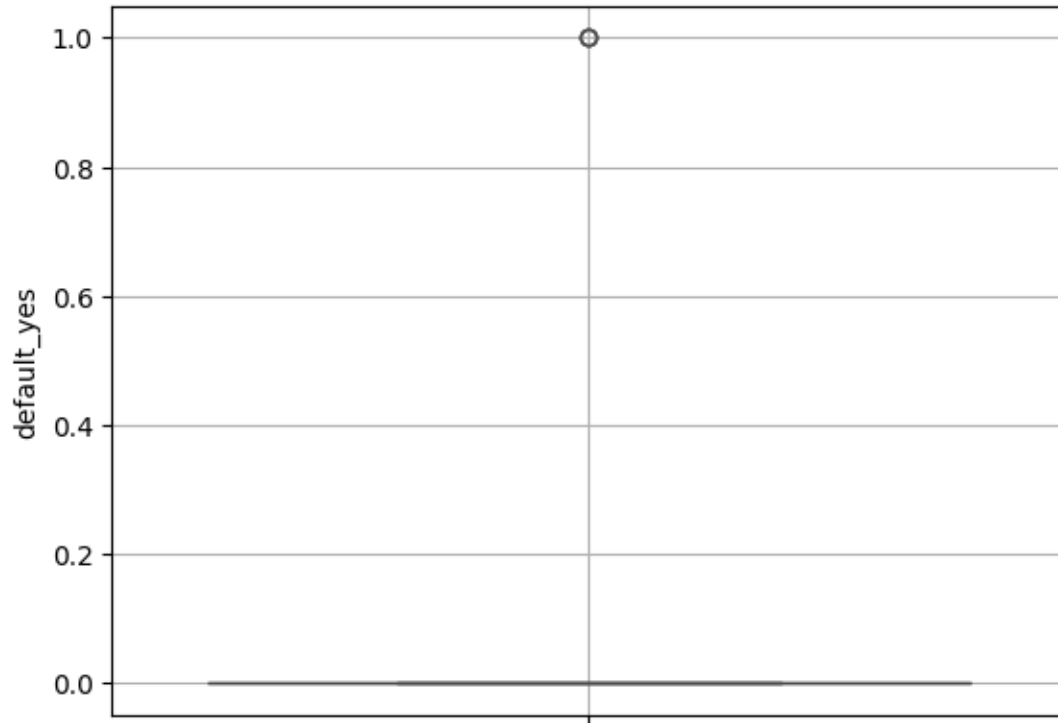
----- default_no -----



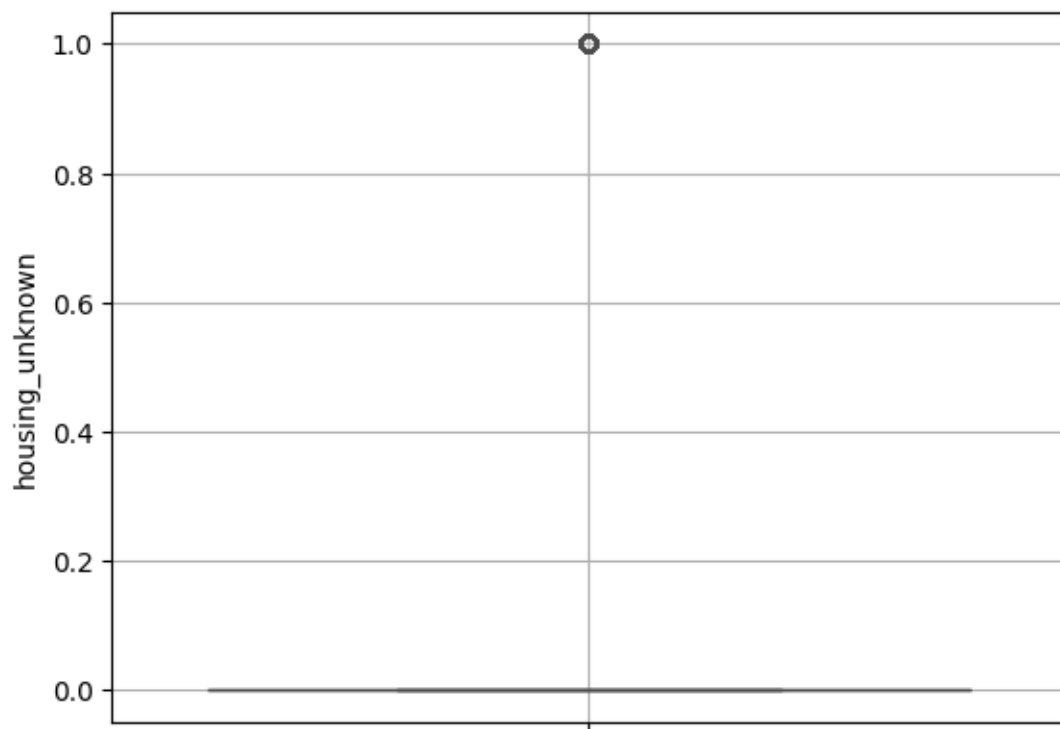
----- default_unknown -----



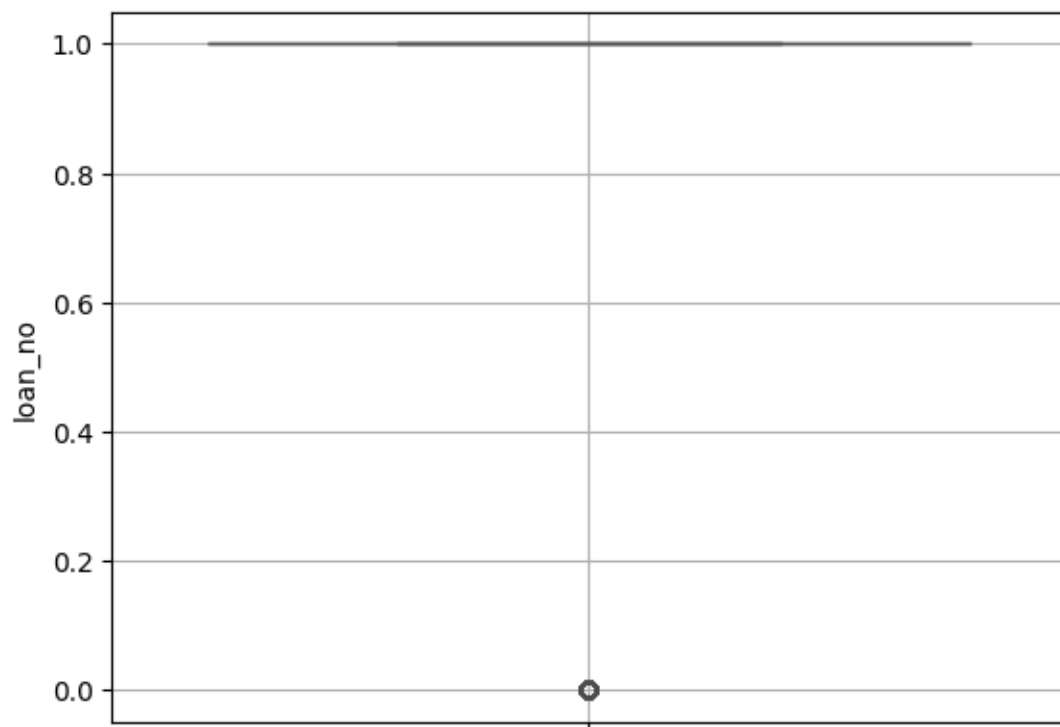
----- default_yes -----



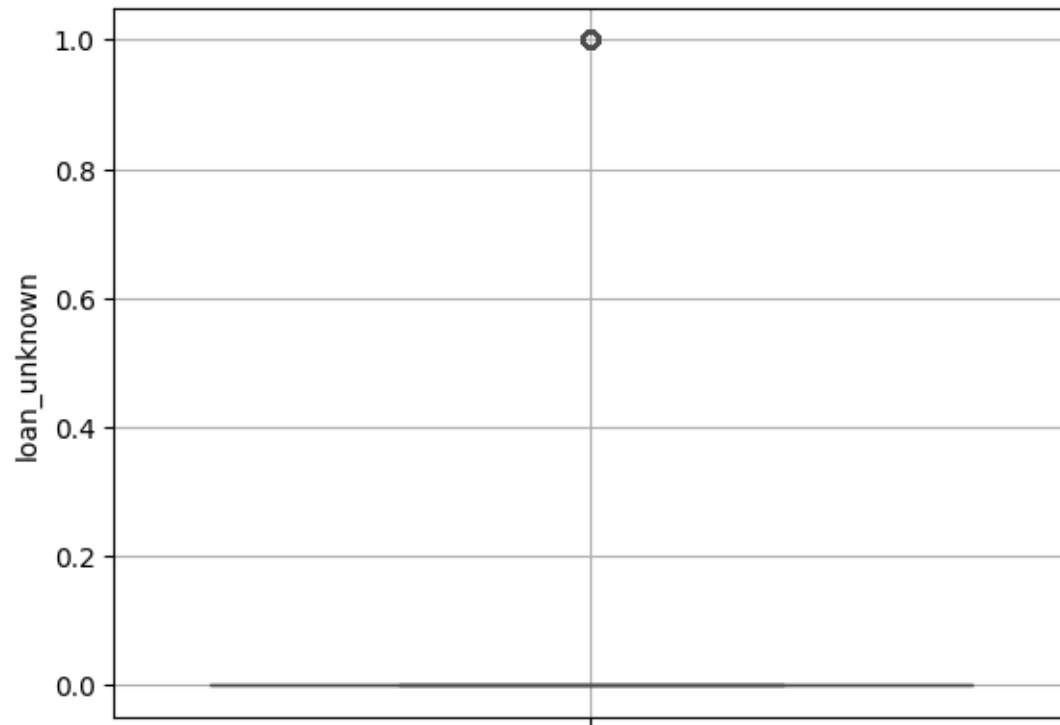
----- housing_unknown -----



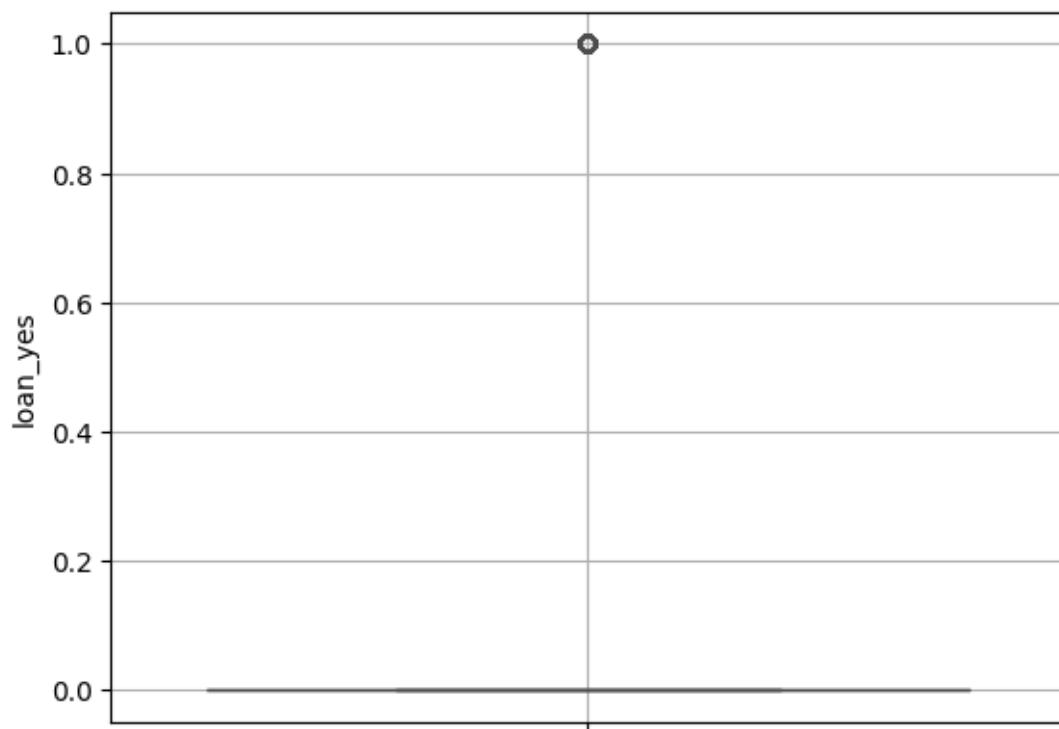
----- loan_no -----



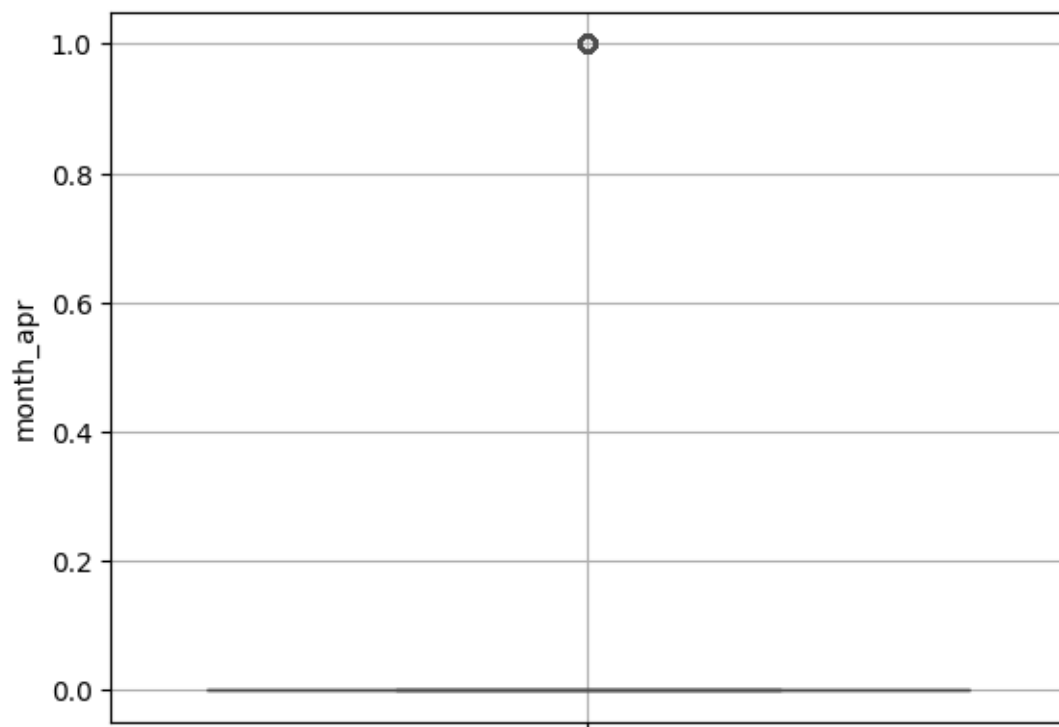
----- loan_unknown -----



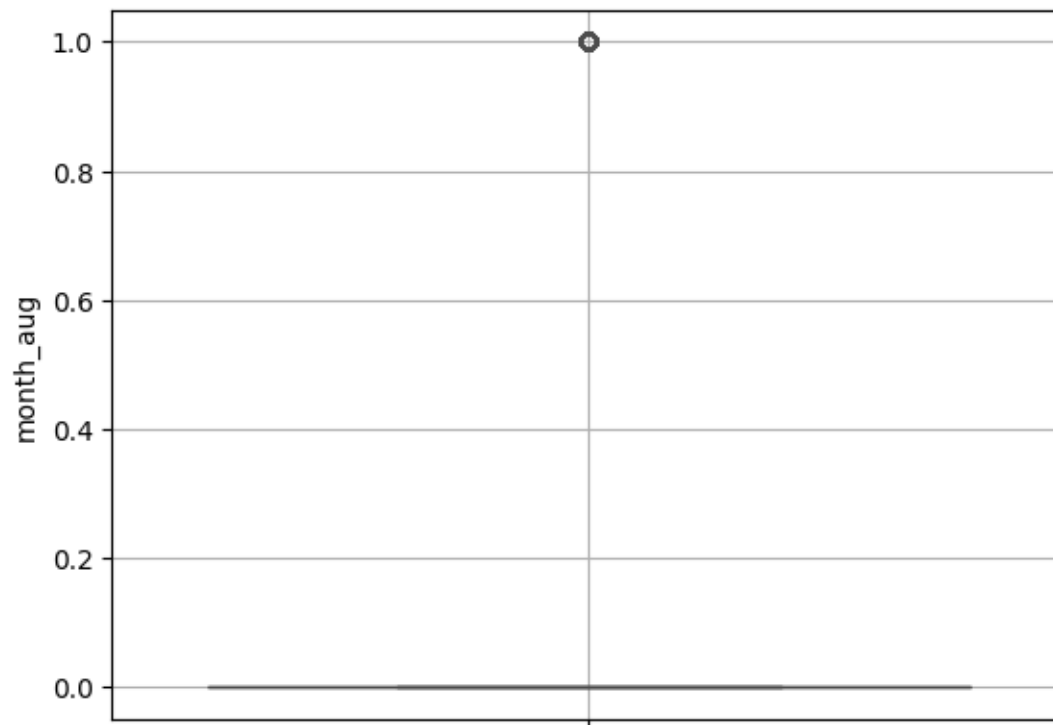
----- loan_yes -----



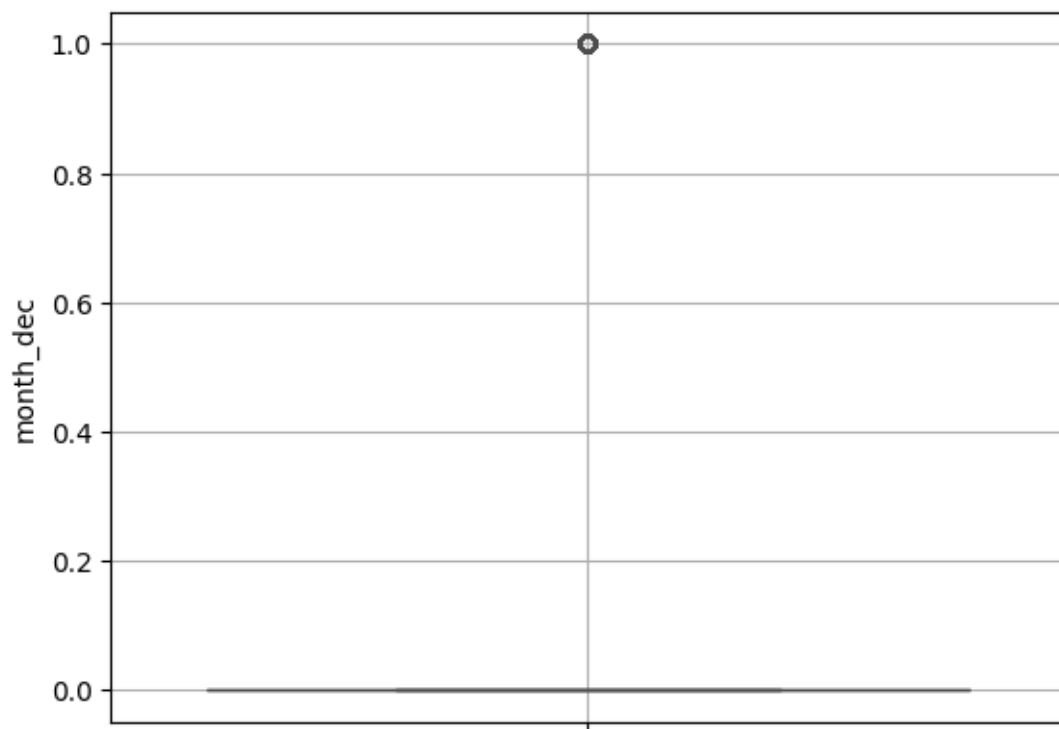
----- month_apr -----



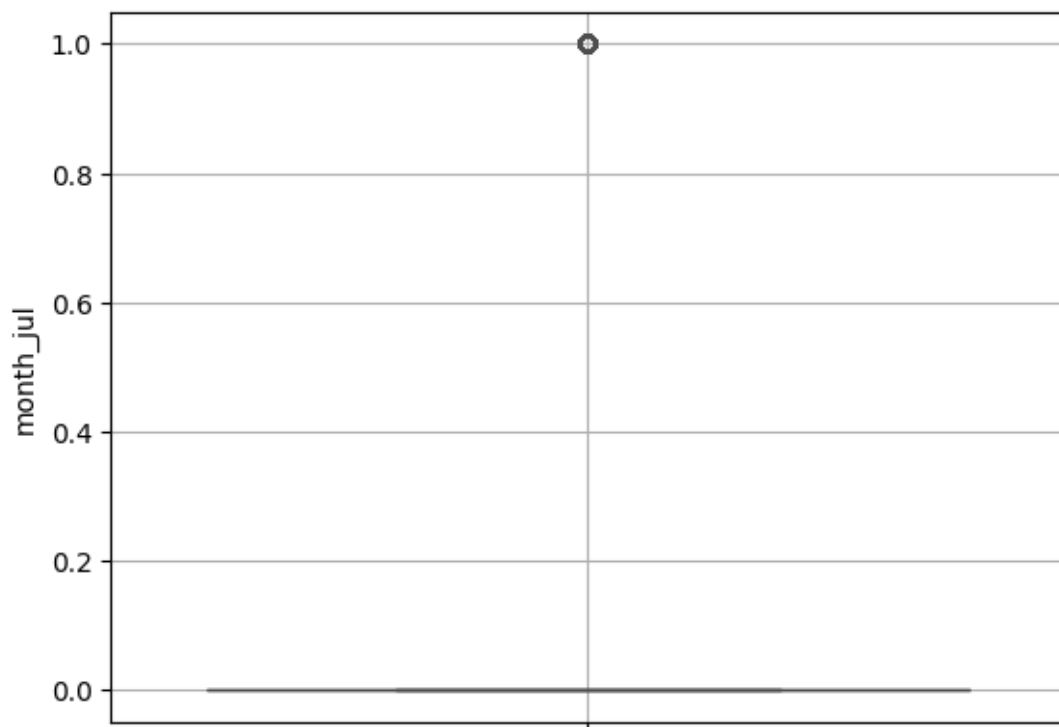
----- month_aug -----



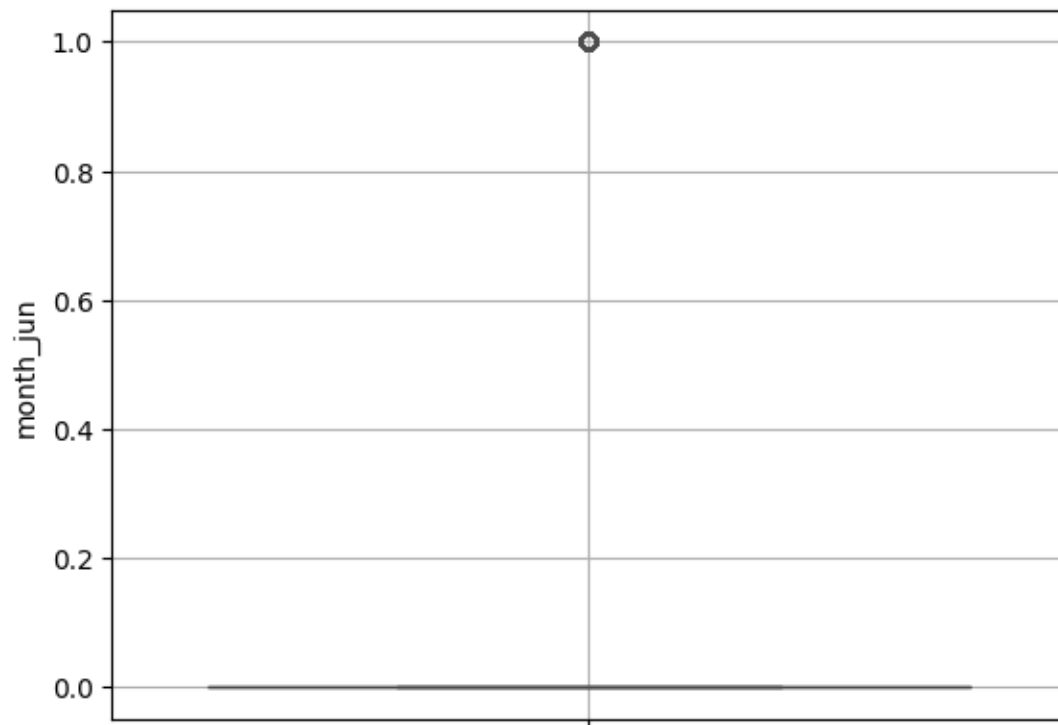
----- month_dec -----



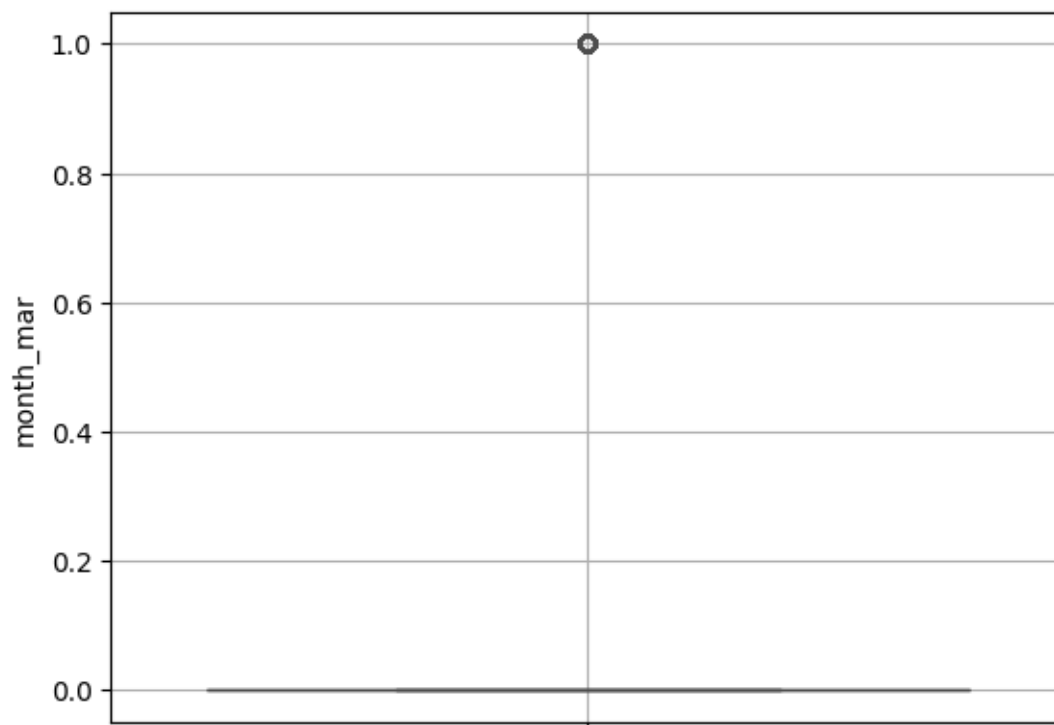
----- month_jul -----



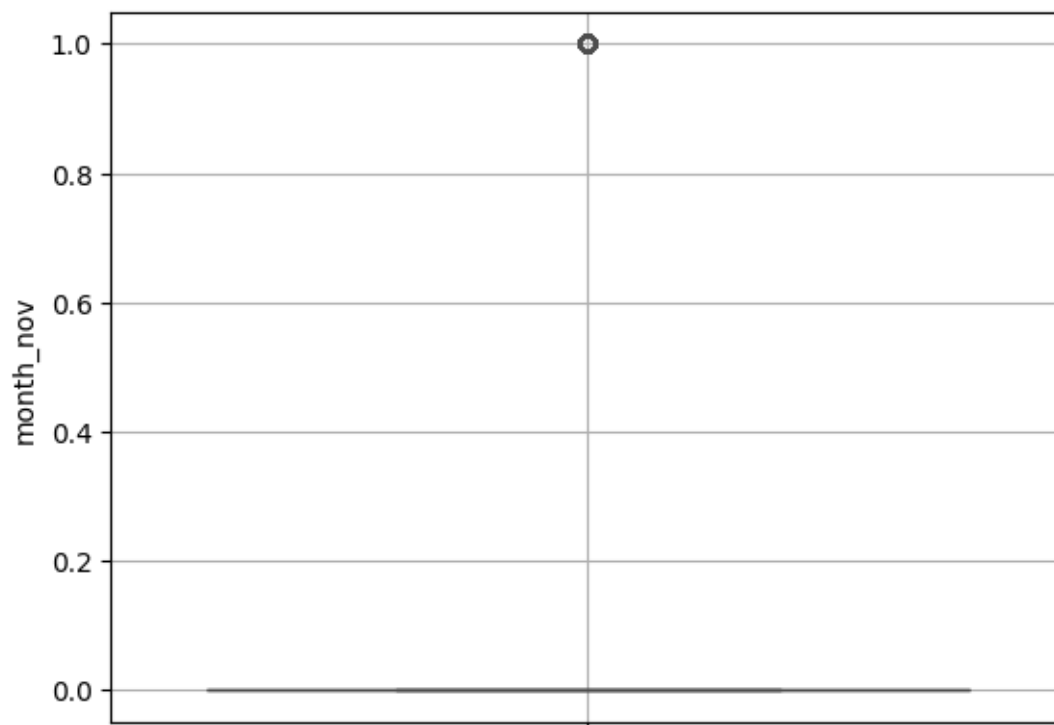
----- month_jun -----

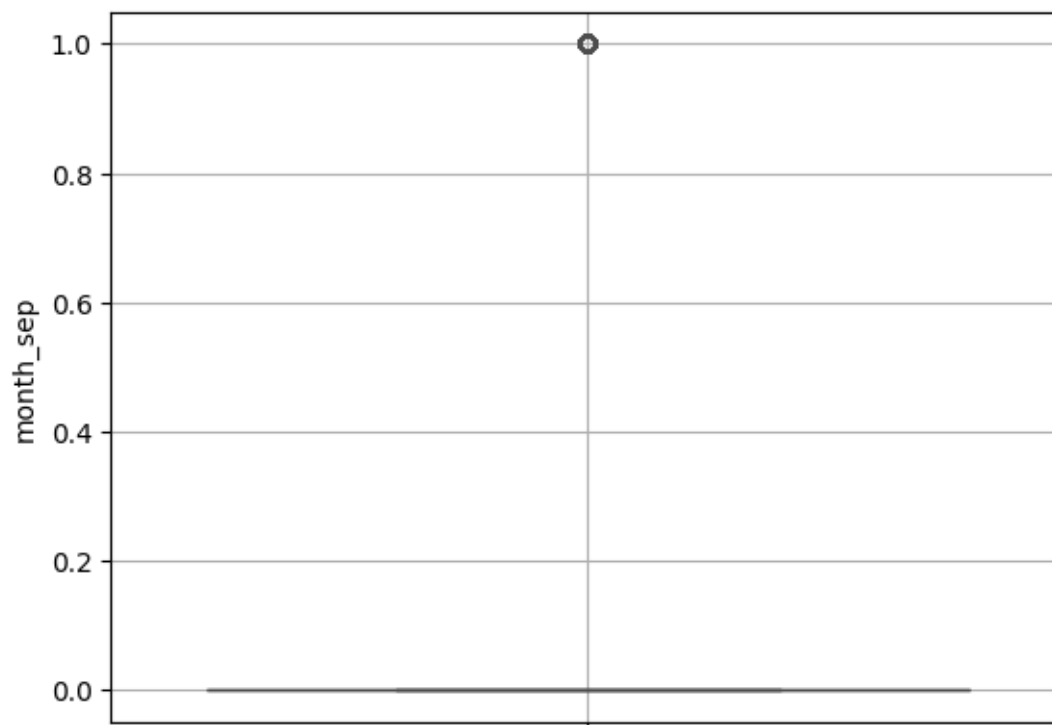


----- month_mar -----

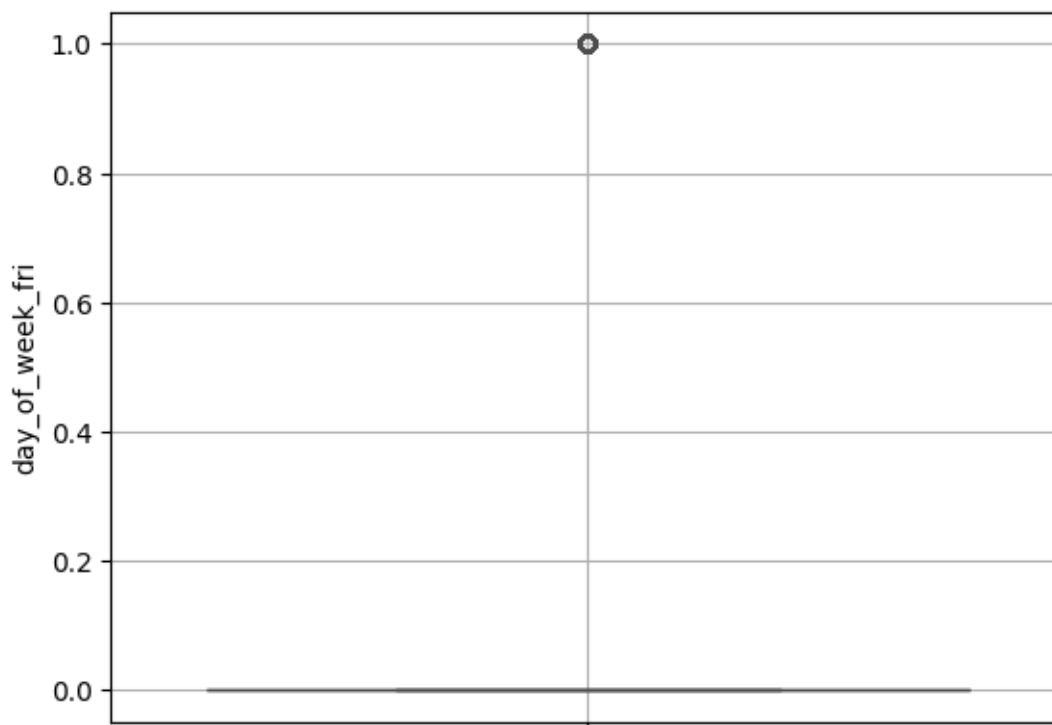


----- month_nov -----

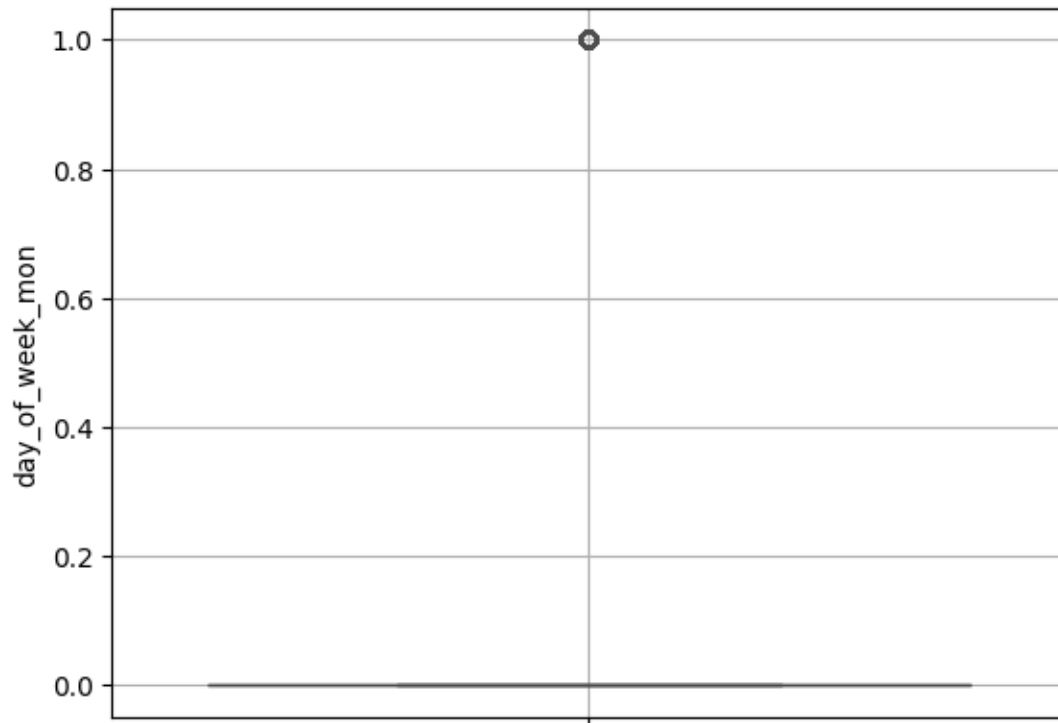




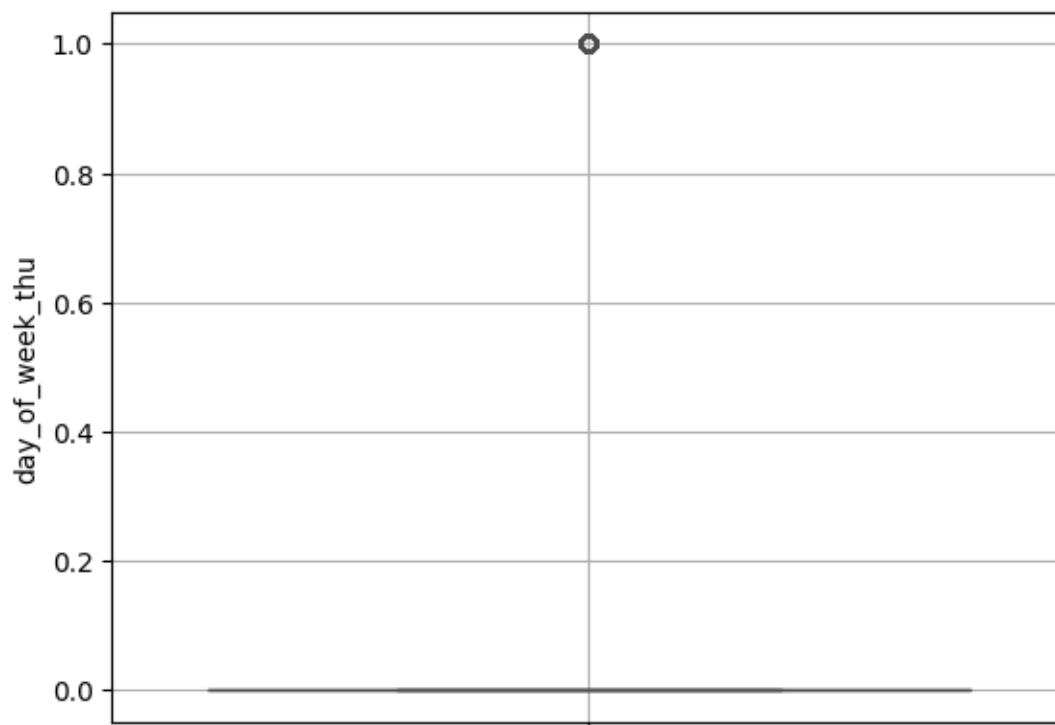
----- day_of_week_fri -----



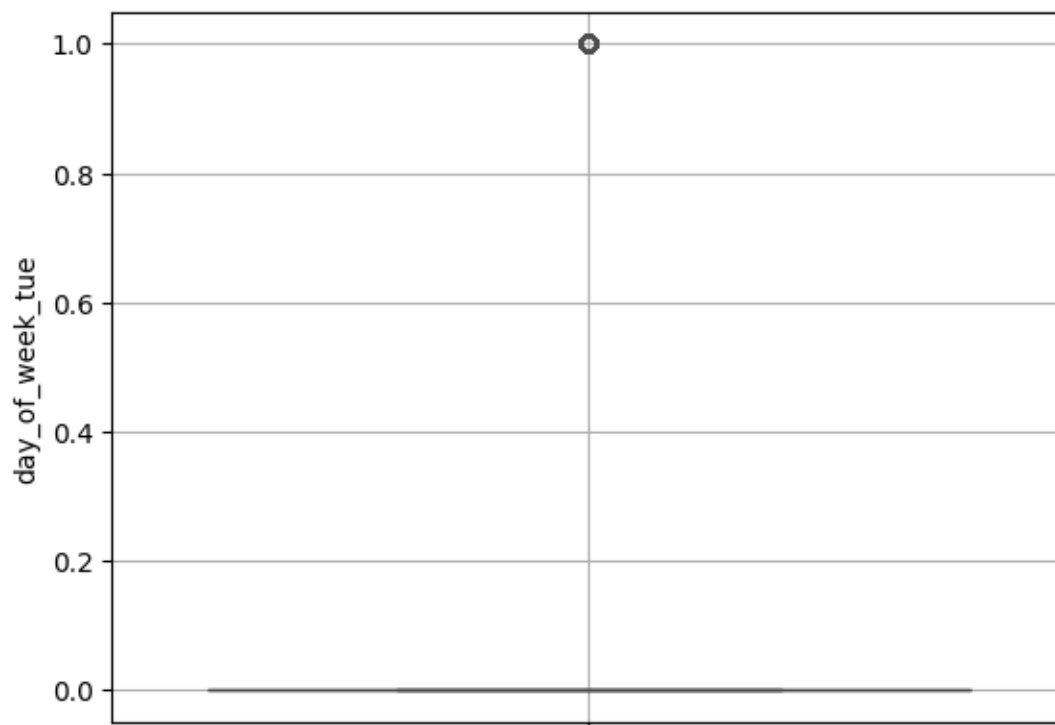
----- day_of_week_mon -----



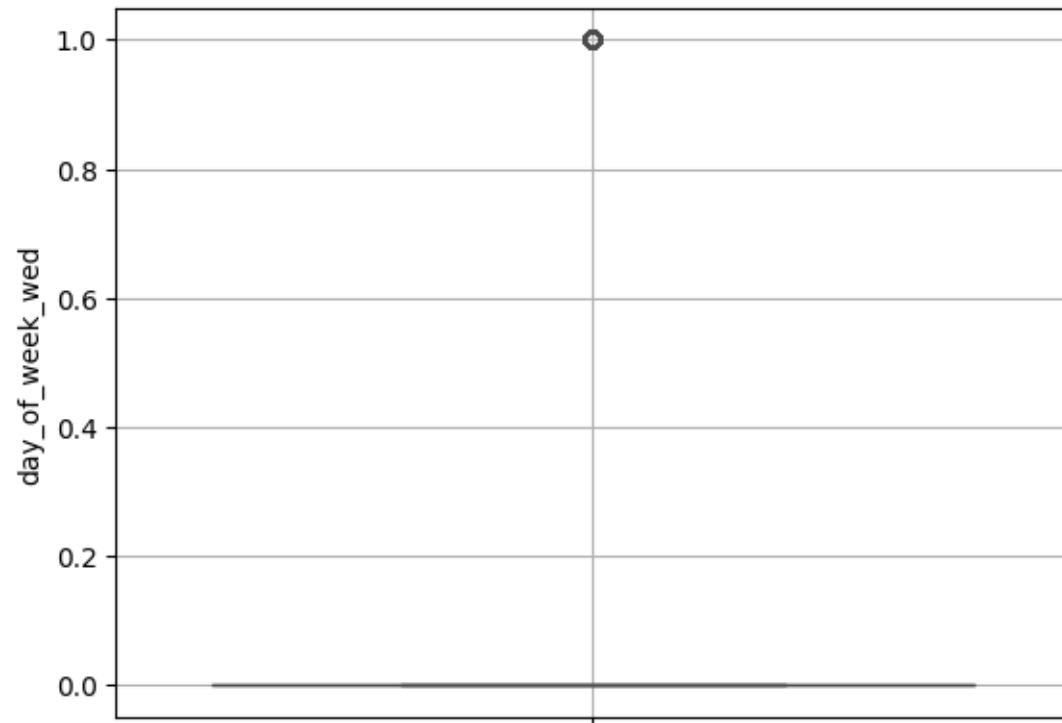
----- day_of_week_thu -----



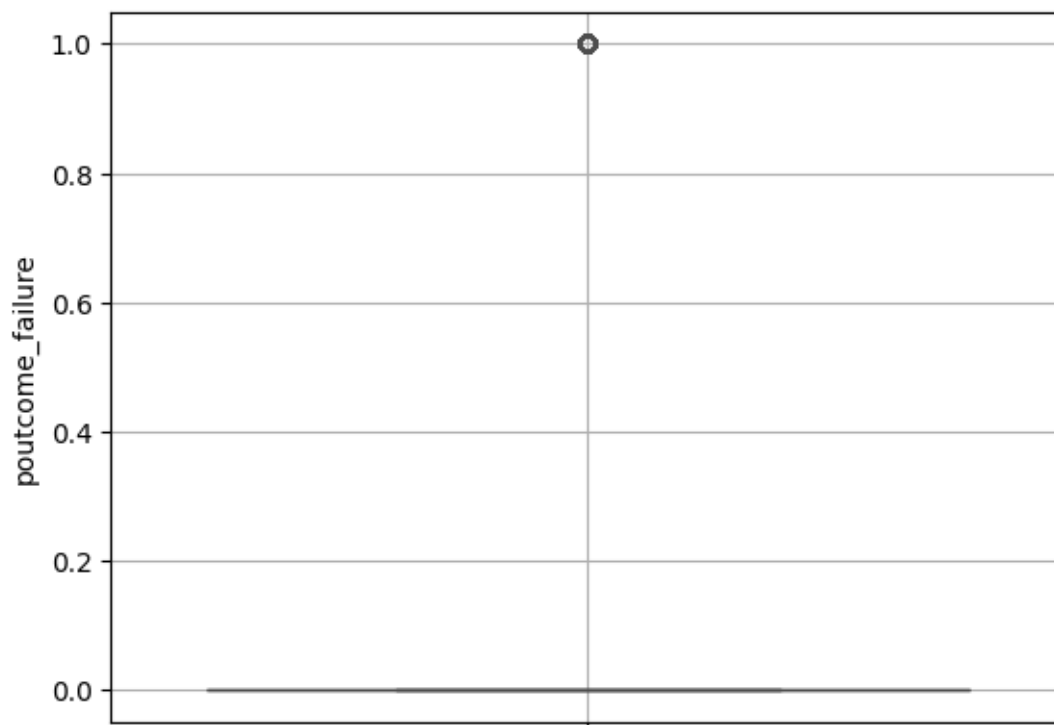
----- `day_of_week_tue` -----



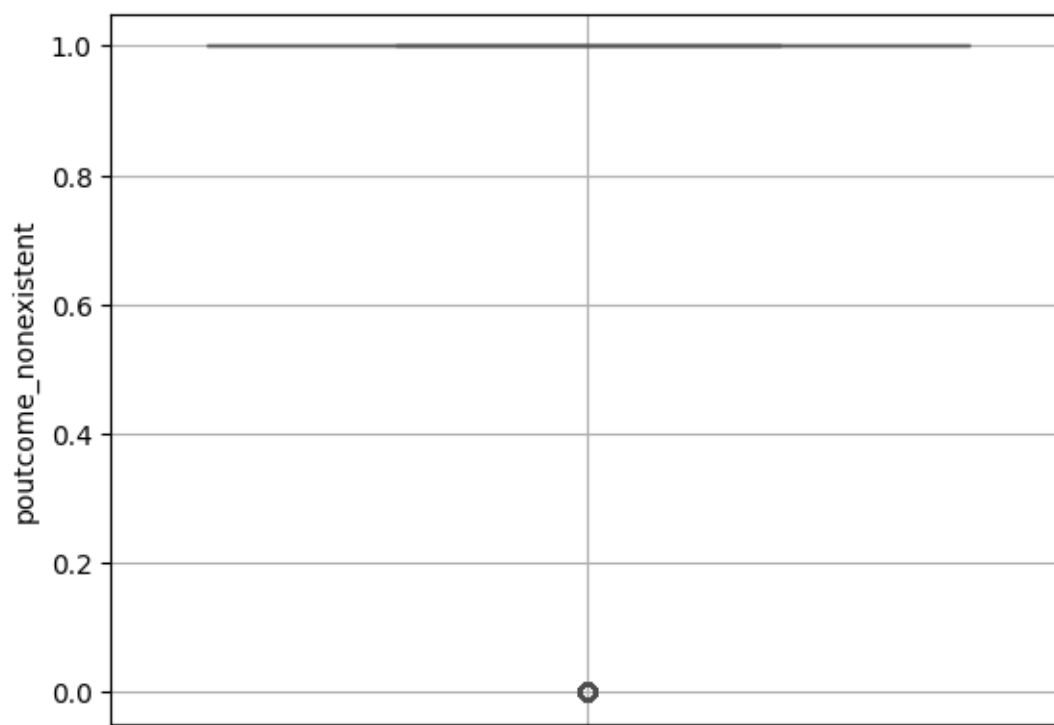
----- day_of_week_wed -----



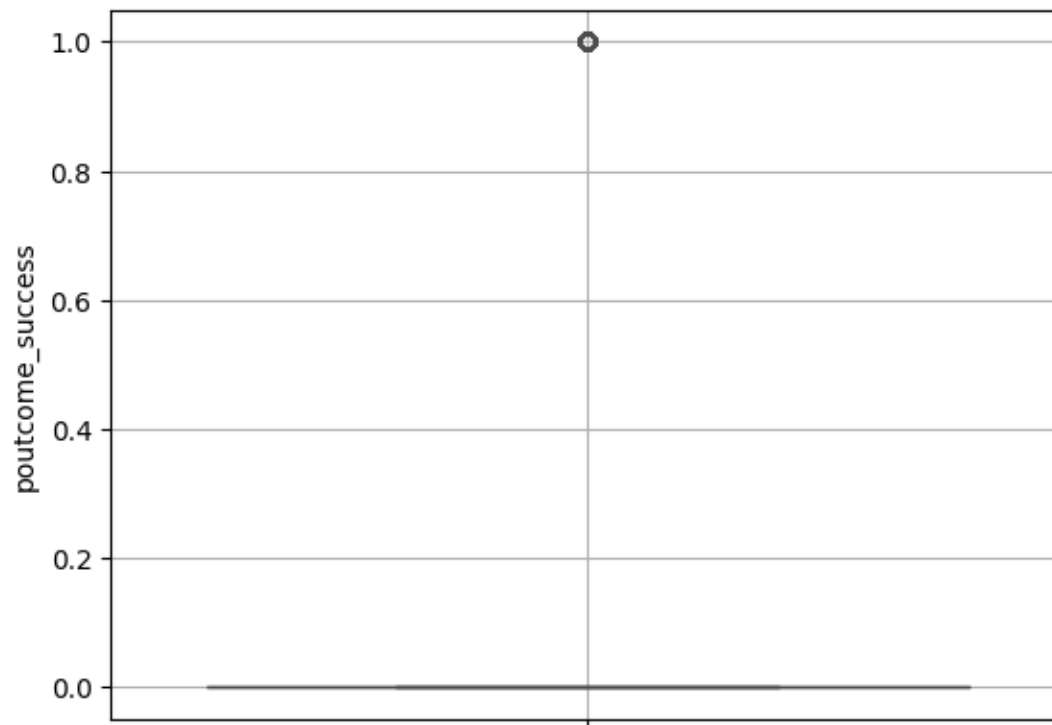
----- poutcome_failure -----



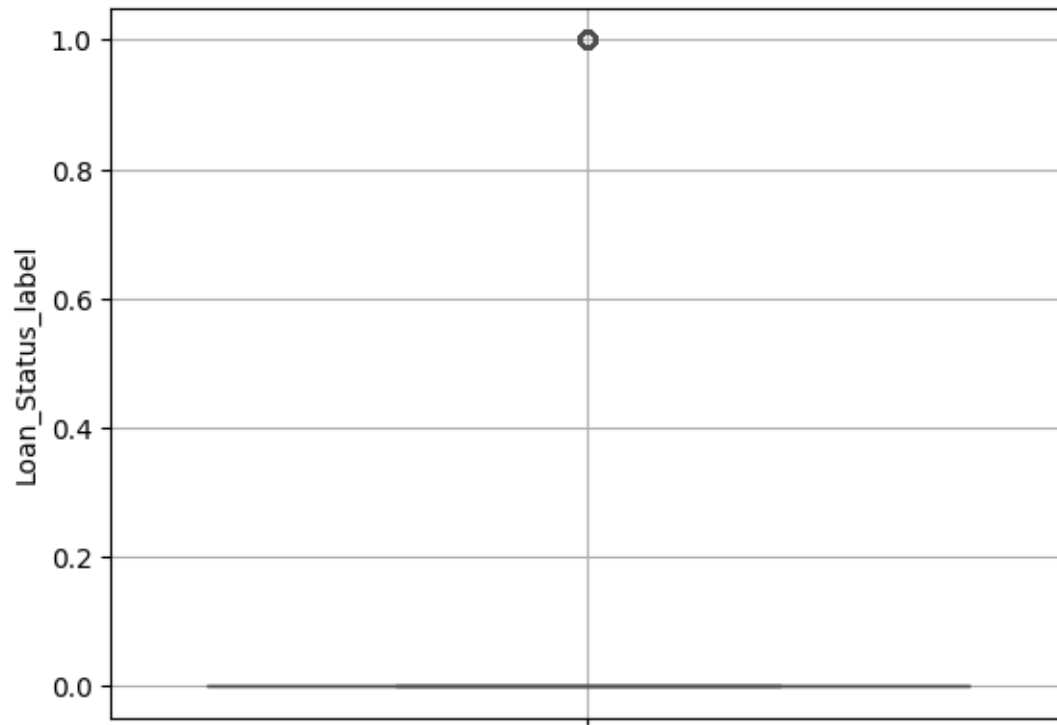
----- poutcome_nonexistent -----



----- poutcome_success -----



----- Loan_Status_label -----



Using IQR (removing outliers)

```
[10]: #IQR

Q1= df.quantile(0.25)
Q3= df.quantile(0.75)

IQR=Q3-Q1
print(f'IQR : \n\n{IQR}')
```

IQR :

age	15.0
campaign	2.0
pdays	0.0
previous	0.0
no_previous_contact	0.0
not_working	0.0
job_admin.	0.0
job_blue-collar	0.0
job_entrepreneur	0.0
job_housemaid	0.0
job_management	0.0
job_retired	0.0

job_self-employed	0.0
job_services	0.0
job_student	0.0
job_technician	0.0
job_unemployed	0.0
job_unknown	0.0
marital_divorced	0.0
marital_married	1.0
marital_single	1.0
marital_unknown	0.0
education_basic.4y	0.0
education_basic.6y	0.0
education_basic.9y	0.0
education_high.school	0.0
education_illiterate	0.0
education_professional.course	0.0
education_university.degree	1.0
education_unknown	0.0
default_no	0.0
default_unknown	0.0
default_yes	0.0
housing_no	1.0
housing_unknown	0.0
housing_yes	1.0
loan_no	0.0
loan_unknown	0.0
loan_yes	0.0
contact_cellular	1.0
contact_telephone	1.0
month_apr	0.0
month_aug	0.0
month_dec	0.0
month_jul	0.0
month_jun	0.0
month_mar	0.0
month_may	1.0
month_nov	0.0
month_oct	0.0
month_sep	0.0
day_of_week_fri	0.0
day_of_week_mon	0.0
day_of_week_thu	0.0
day_of_week_tue	0.0
day_of_week_wed	0.0
poutcome_failure	0.0
poutcome_nonexistent	0.0
poutcome_success	0.0
Loan_Status_label	0.0

dtype: float64

```
[11]: lower= Q1 - 1.5*IQR
      upper= Q3 + 1.5*IQR

      print(f'Lower bound for outliers: \n\n{lower}\n\n')
      print(f'Upper bound for outliers: \n\n{upper}')
```

Lower bound for outliers:

age	9.5
campaign	-2.0
pdays	999.0
previous	0.0
no_previous_contact	1.0
not_working	0.0
job_admin.	0.0
job_blue-collar	0.0
job_entrepreneur	0.0
job_housemaid	0.0
job_management	0.0
job_retired	0.0
job_self-employed	0.0
job_services	0.0
job_student	0.0
job_technician	0.0
job_unemployed	0.0
job_unknown	0.0
marital_divorced	0.0
marital_married	-1.5
marital_single	-1.5
marital_unknown	0.0
education_basic.4y	0.0
education_basic.6y	0.0
education_basic.9y	0.0
education_high.school	0.0
education_illiterate	0.0
education_professional.course	0.0
education_university.degree	-1.5
education_unknown	0.0
default_no	1.0
default_unknown	0.0
default_yes	0.0
housing_no	-1.5
housing_unknown	0.0
housing_yes	-1.5
loan_no	1.0
loan_unknown	0.0

loan_yes	0.0
contact_cellular	-1.5
contact_telephone	-1.5
month_apr	0.0
month_aug	0.0
month_dec	0.0
month_jul	0.0
month_jun	0.0
month_mar	0.0
month_may	-1.5
month_nov	0.0
month_oct	0.0
month_sep	0.0
day_of_week_fri	0.0
day_of_week_mon	0.0
day_of_week_thu	0.0
day_of_week_tue	0.0
day_of_week_wed	0.0
poutcome_failure	0.0
poutcome_nonexistent	1.0
poutcome_success	0.0
Loan_Status_label	0.0
dtype: float64	

Upper bound for outliers:

age	69.5
campaign	6.0
pdays	999.0
previous	0.0
no_previous_contact	1.0
not_working	0.0
job_admin.	0.0
job_blue-collar	0.0
job_entrepreneur	0.0
job_housemaid	0.0
job_management	0.0
job_retired	0.0
job_self-employed	0.0
job_services	0.0
job_student	0.0
job_technician	0.0
job_unemployed	0.0
job_unknown	0.0
marital_divorced	0.0
marital_married	2.5
marital_single	2.5

marital_unknown	0.0
education_basic.4y	0.0
education_basic.6y	0.0
education_basic.9y	0.0
education_high.school	0.0
education_illiterate	0.0
education_professional.course	0.0
education_university.degree	2.5
education_unknown	0.0
default_no	1.0
default_unknown	0.0
default_yes	0.0
housing_no	2.5
housing_unknown	0.0
housing_yes	2.5
loan_no	1.0
loan_unknown	0.0
loan_yes	0.0
contact_cellular	2.5
contact_telephone	2.5
month_apr	0.0
month_aug	0.0
month_dec	0.0
month_jul	0.0
month_jun	0.0
month_mar	0.0
month_may	2.5
month_nov	0.0
month_oct	0.0
month_sep	0.0
day_of_week_fri	0.0
day_of_week_mon	0.0
day_of_week_thu	0.0
day_of_week_tue	0.0
day_of_week_wed	0.0
poutcome_failure	0.0
poutcome_nonexistent	1.0
poutcome_success	0.0
Loan_Status_label	0.0
dtype: float64	

```
[12]: df.shape
```

```
[12]: (38771, 60)
```

```
[13]: df[((df<lower) | (df>upper)).any(axis=1)]
```

```
[13]:      age  campaign  pdays  previous  no_previous_contact  not_working  \
0      56      1      999      0      1      0
1      57      1      999      0      1      0
2      37      1      999      0      1      0
3      40      1      999      0      1      0
4      56      1      999      0      1      0
...  ...
41183  73      1      999      0      1      1
41184  46      1      999      0      1      0
41185  56      2      999      0      1      1
41186  44      1      999      0      1      0
41187  74      3      999      1      1      1
```

```
      job_admin.  job_blue-collar  job_entrepreneur  job_housemaid  ...  \
0      0      0      0      1  ...
1      0      0      0      0  ...
2      0      0      0      0  ...
3      1      0      0      0  ...
4      0      0      0      0  ...
...  ...
41183      0      0      0      0  ...
41184      0      1      0      0  ...
41185      0      0      0      0  ...
41186      0      0      0      0  ...
41187      0      0      0      0  ...
```

```
      month_sep  day_of_week_fri  day_of_week_mon  day_of_week_thu  \
0      0      0      1      0
1      0      0      1      0
2      0      0      1      0
3      0      0      1      0
4      0      0      1      0
...  ...
41183      0      1      0      0
41184      0      1      0      0
41185      0      1      0      0
41186      0      1      0      0
41187      0      1      0      0
```

```
      day_of_week_tue  day_of_week_wed  poutcome_failure  \
0      0      0      0
1      0      0      0
2      0      0      0
3      0      0      0
4      0      0      0
...  ...
41183      0      0      0
```

41184	0	0	0
41185	0	0	0
41186	0	0	0
41187	0	0	1

	poutcome_nonexistent	poutcome_success	Loan_Status_label
0	1	0	0
1	1	0	0
2	1	0	0
3	1	0	0
4	1	0	0
...	...	...	...
41183	1	0	1
41184	1	0	0
41185	1	0	0
41186	1	0	1
41187	0	0	0

[38771 rows x 60 columns]

```
[37]: df[~((df<lower)| (df>upper)).any(axis=1)]
```

[37]: Empty DataFrame

Columns: [age, campaign, pdays, previous, no_previous_contact, not_working, job_admin., job_blue-collar, job_entrepreneur, job_housemaid, job_management, job_retired, job_self-employed, job_services, job_student, job_technician, job_unemployed, job_unknown, marital_divorced, marital_married, marital_single, marital_unknown, education_basic.4y, education_basic.6y, education_basic.9y, education_high.school, education_illiterate, education_professional.course, education_university.degree, education_unknown, default_no, default_unknown, default_yes, housing_no, housing_unknown, housing_yes, loan_no, loan_unknown, loan_yes, contact_cellular, contact_telephone, month_apr, month_aug, month_dec, month_jul, month_jun, month_mar, month_may, month_nov, month_oct, month_sep, day_of_week_fri, day_of_week_mon, day_of_week_thu, day_of_week_tue, day_of_week_wed, poutcome_failure, poutcome_nonexistent, poutcome_success, Loan_Status_label]
Index: []

[0 rows x 60 columns]

```
[14]: df
```

[14]:	age	campaign	pdays	previous	no_previous_contact	not_working	\
0	56	1	999	0	1	0	
1	57	1	999	0	1	0	
2	37	1	999	0	1	0	
3	40	1	999	0	1	0	

4	56	1	999	0	1	0
...	...	...	...	...	...	...
41183	73	1	999	0	1	1
41184	46	1	999	0	1	0
41185	56	2	999	0	1	1
41186	44	1	999	0	1	0
41187	74	3	999	1	1	1

	job_admin.	job_blue-collar	job_entrepreneur	job_housemaid	...	\
0	0	0	0	1	...	
1	0	0	0	0	...	
2	0	0	0	0	...	
3	1	0	0	0	...	
4	0	0	0	0	...	
...	...	...	...	...	...	
41183	0	0	0	0	...	
41184	0	1	0	0	...	
41185	0	0	0	0	...	
41186	0	0	0	0	...	
41187	0	0	0	0	...	

	month_sep	day_of_week_fri	day_of_week_mon	day_of_week_thu	\
0	0	0	1	0	
1	0	0	1	0	
2	0	0	1	0	
3	0	0	1	0	
4	0	0	1	0	
...	...	...	...	...	
41183	0	1	0	0	
41184	0	1	0	0	
41185	0	1	0	0	
41186	0	1	0	0	
41187	0	1	0	0	

	day_of_week_tue	day_of_week_wed	poutcome_failure	\
0	0	0	0	
1	0	0	0	
2	0	0	0	
3	0	0	0	
4	0	0	0	
...	...	...	...	
41183	0	0	0	
41184	0	0	0	
41185	0	0	0	
41186	0	0	0	
41187	0	0	1	

	poutcome_nonexistent	poutcome_success	Loan_Status_label
0	1	0	0
1	1	0	0
2	1	0	0
3	1	0	0
4	1	0	0
...	...	...	...
41183	1	0	1
41184	1	0	0
41185	1	0	0
41186	1	0	1
41187	0	0	0

[38771 rows x 60 columns]

Feature selection

```
[15]: df.corr()
```

```
[15]:
```

	age	campaign	pdays	previous	\
age	1.000000	-0.000792	-0.032680	0.020430	
campaign	-0.000792	1.000000	0.057855	-0.086774	
pdays	-0.032680	0.057855	1.000000	-0.588940	
previous	0.020430	-0.086774	-0.588940	1.000000	
no_previous_contact	-0.032600	0.057840	0.999992	-0.588887	
not_working	0.207835	-0.021846	-0.112069	0.103602	
job_admin.	-0.095507	0.015630	-0.028564	0.023149	
job_blue-collar	-0.030068	0.000038	0.066963	-0.056865	
job_entrepreneur	0.027431	-0.004086	0.020523	-0.015668	
job_housemaid	0.084198	0.002921	-0.000918	-0.012511	
job_management	0.059189	-0.011188	0.001119	0.005845	
job_retired	0.443493	-0.009917	-0.071670	0.064722	
job_self-employed	-0.003106	0.004022	0.015712	-0.012723	
job_services	-0.067009	0.002252	0.031735	-0.012832	
job_student	-0.203179	-0.026524	-0.096310	0.105895	
job_technician	-0.064292	0.003133	0.003637	-0.016419	
job_unemployed	-0.005351	-0.002383	-0.022260	0.006921	
job_unknown	0.046759	0.001406	-0.010792	-0.004902	
marital_divorced	0.162918	0.002671	0.013475	-0.005370	
marital_married	0.269629	0.002627	0.032378	-0.045565	
marital_single	-0.409824	-0.005692	-0.044312	0.052630	
marital_unknown	0.001062	0.009246	-0.005616	0.008662	
education_basic.4y	0.242494	0.004262	0.006228	-0.017857	
education_basic.6y	0.006994	-0.002879	0.024657	-0.022823	
education_basic.9y	-0.038092	-0.004963	0.038438	-0.028995	
education_high.school	-0.109122	-0.000994	0.006236	0.013670	
education_illiterate	0.017108	-0.002723	-0.001836	-0.002989	
education_professional.course	0.004070	0.002892	-0.003901	-0.008161	

education_university.degree	-0.067088	0.001510	-0.040012	0.030187
education_unknown	0.065671	-0.000185	-0.021068	0.020728
default_no	-0.159712	-0.032852	-0.083360	0.107046
default_unknown	0.159694	0.032943	0.083333	-0.107117
default_yes	0.001813	-0.004045	0.001773	0.002639
housing_no	0.000434	0.010616	0.010893	-0.022029
housing_unknown	-0.002609	-0.003557	0.003589	0.001115
housing_yes	0.000387	-0.009461	-0.011982	0.021602
loan_no	0.011931	0.003900	-0.004977	0.006929
loan_unknown	-0.002609	-0.003557	0.003589	0.001115
loan_yes	-0.011530	-0.002608	0.003737	-0.007826
contact_cellular	-0.000853	-0.073606	-0.124154	0.221757
contact_telephone	0.000853	0.073606	0.124154	-0.221757
month_apr	0.014849	-0.061498	-0.008864	0.077544
month_aug	0.074766	0.024229	-0.007298	-0.042323
month_dec	0.047459	-0.011496	-0.076032	0.059138
month_jul	-0.041768	0.105561	0.049869	-0.116840
month_jun	-0.012368	0.066985	0.018597	-0.074076
month_mar	0.006080	-0.016941	-0.079622	0.073909
month_may	-0.072065	-0.036134	0.073731	-0.017015
month_nov	0.029354	-0.081776	-0.016773	0.081364
month_oct	0.049442	-0.051951	-0.128860	0.126791
month_sep	0.040026	-0.036927	-0.154564	0.157224
day_of_week_fri	0.006287	0.025805	0.014940	0.003027
day_of_week_mon	0.017340	0.010532	0.002517	-0.004078
day_of_week_thu	-0.019094	0.007932	-0.009324	0.002367
day_of_week_tue	0.017902	-0.024070	-0.008213	0.001799
day_of_week_wed	-0.022350	-0.020438	0.000339	-0.003039
poutcome_failure	-0.007085	-0.075279	0.009098	0.678092
poutcome_nonexistent	-0.011749	0.096142	0.493971	-0.877385
poutcome_success	0.034118	-0.055900	-0.950559	0.525045
Loan_Status_label	0.026865	-0.074539	-0.324611	0.228665

	no_previous_contact	not_working	job_admin.	\
age	-0.032600	0.207835	-0.095507	
campaign	0.057840	-0.021846	0.015630	
pdays	0.999992	-0.112069	-0.028564	
previous	-0.588887	0.103602	0.023149	
no_previous_contact	1.000000	-0.112069	-0.028578	
not_working	-0.112069	1.000000	-0.181604	
job_admin.	-0.028578	-0.181604	1.000000	
job_blue-collar	0.066956	-0.169473	-0.308601	
job_entrepreneur	0.020524	-0.061192	-0.111427	
job_housemaid	-0.000907	-0.051962	-0.094620	
job_management	0.001144	-0.088057	-0.160348	
job_retired	-0.071633	0.673497	-0.122310	
job_self-employed	0.015730	-0.060669	-0.110476	

job_services	0.031757	-0.103370	-0.188231
job_student	-0.096389	0.473514	-0.085992
job_technician	0.003601	-0.138747	-0.252651
job_unemployed	-0.022234	0.512586	-0.093088
job_unknown	-0.010747	-0.028990	-0.052790
marital_divorced	0.013479	0.018261	0.024301
marital_married	0.032410	-0.039913	-0.115414
marital_single	-0.044346	0.029790	0.109354
marital_unknown	-0.005655	0.007640	-0.007455
education_basic.4y	0.006238	0.104831	-0.179023
education_basic.6y	0.024661	-0.031976	-0.104327
education_basic.9y	0.038416	-0.023543	-0.155420
education_high.school	0.006218	0.011312	0.124887
education_illiterate	-0.001836	0.005702	-0.009623
education_professional.course	-0.003950	-0.006794	-0.157373
education_university.degree	-0.039926	-0.065947	0.315790
education_unknown	-0.021123	0.055758	-0.051269
default_no	-0.083364	0.005923	0.119956
default_unknown	0.083337	-0.006084	-0.119863
default_yes	0.001773	0.007433	-0.005059
housing_no	0.010859	-0.007070	-0.008045
housing_unknown	0.003590	0.002415	-0.007339
housing_yes	-0.011949	0.006286	0.010322
loan_no	-0.004997	0.005532	-0.018850
loan_unknown	0.003590	0.002415	-0.007339
loan_yes	0.003758	-0.006903	0.023138
contact_cellular	-0.124149	0.044449	0.055031
contact_telephone	0.124149	-0.044449	-0.055031
month_apr	-0.008826	0.029452	-0.001388
month_aug	-0.007286	0.015076	0.066101
month_dec	-0.076045	0.060065	0.004395
month_jul	0.049784	-0.021448	0.000376
month_jun	0.018646	-0.006642	-0.016832
month_mar	-0.079706	0.055113	0.016884
month_may	0.073710	-0.071555	-0.051287
month_nov	-0.016691	-0.004681	0.007106
month_oct	-0.128888	0.092685	0.009174
month_sep	-0.154550	0.084659	0.011884
day_of_week_fri	0.014932	0.001643	0.007820
day_of_week_mon	0.002493	-0.006026	-0.000429
day_of_week_thu	-0.009325	-0.001012	-0.003825
day_of_week_tue	-0.008194	0.007139	-0.000034
day_of_week_wed	0.000353	-0.001549	-0.003369
poutcome_failure	0.008844	0.031470	0.005870
poutcome_nonexistent	0.493975	-0.082430	-0.020046
poutcome_success	-0.950141	0.103344	0.028119
Loan_Status_label	-0.324572	0.118294	0.035250

	job_blue-collar	job_entrepreneur \
age	-0.030068	0.027431
campaign	0.000038	-0.004086
pdays	0.066963	0.020523
previous	-0.056865	-0.015668
no_previous_contact	0.066956	0.020524
not_working	-0.169473	-0.061192
job_admin.	-0.308601	-0.111427
job_blue-collar	1.000000	-0.103984
job_entrepreneur	-0.103984	1.000000
job_housemaid	-0.088300	-0.031882
job_management	-0.149637	-0.054029
job_retired	-0.114140	-0.041212
job_self-employed	-0.103096	-0.037225
job_services	-0.175657	-0.063425
job_student	-0.080248	-0.028975
job_technician	-0.235774	-0.085131
job_unemployed	-0.086870	-0.031366
job_unknown	-0.049264	-0.017788
marital_divorced	-0.056566	0.004979
marital_married	0.123230	0.052030
marital_single	-0.093710	-0.060318
marital_unknown	-0.005030	0.000433
education_basic.4y	0.260215	-0.004962
education_basic.6y	0.232231	-0.005099
education_basic.9y	0.359995	-0.000310
education_high.school	-0.170027	-0.033990
education_illiterate	0.011424	0.008646
education_professional.course	-0.122760	-0.021177
education_university.degree	-0.331890	0.053987
education_unknown	0.017294	-0.002621
default_no	-0.172466	0.001707
default_unknown	0.172591	-0.001671
default_yes	-0.004721	-0.001705
housing_no	0.015215	-0.004531
housing_unknown	0.007142	-0.000376
housing_yes	-0.017405	0.004634
loan_no	0.001103	0.006815
loan_unknown	0.007142	-0.000376
loan_yes	-0.004236	-0.007065
contact_cellular	-0.091551	-0.019875
contact_telephone	0.091551	0.019875
month_apr	-0.006973	0.007606
month_aug	-0.124648	-0.050576
month_dec	-0.025855	-0.011244
month_jul	0.025237	0.005953

month_jun	0.032399	0.011466
month_mar	-0.036885	-0.020385
month_may	0.131799	-0.000362
month_nov	-0.060306	0.051826
month_oct	-0.049936	-0.010895
month_sep	-0.055378	-0.009700
day_of_week_fri	0.002939	-0.000656
day_of_week_mon	-0.009880	0.005297
day_of_week_thu	-0.007824	0.004588
day_of_week_tue	-0.004312	-0.005552
day_of_week_wed	0.019503	-0.003921
poutcome_failure	-0.014362	-0.000734
poutcome_nonexistent	0.045937	0.010381
poutcome_success	-0.062914	-0.018425
Loan_Status_label	-0.075625	-0.019306

	job_housemaid	...	month_sep	day_of_week_fri	\
age	0.084198	...	0.040026	0.006287	
campaign	0.002921	...	-0.036927	0.025805	
pdays	-0.000918	...	-0.154564	0.014940	
previous	-0.012511	...	0.157224	0.003027	
no_previous_contact	-0.000907	...	-0.154550	0.014932	
not_working	-0.051962	...	0.084659	0.001643	
job_admin.	-0.094620	...	0.011884	0.007820	
job_blue-collar	-0.088300	...	-0.055378	0.002939	
job_entrepreneur	-0.031882	...	-0.009700	-0.000656	
job_housemaid	1.000000	...	-0.003855	-0.007712	
job_management	-0.045880	...	0.003590	-0.005322	
job_retired	-0.034996	...	0.064144	0.002071	
job_self-employed	-0.031610	...	-0.000087	0.003728	
job_services	-0.053858	...	-0.018639	0.001324	
job_student	-0.024605	...	0.053999	-0.002020	
job_technician	-0.072291	...	-0.000148	-0.004830	
job_unemployed	-0.026635	...	0.021302	0.002193	
job_unknown	-0.015105	...	0.007778	-0.018062	
marital_divorced	0.020771	...	-0.009833	-0.005992	
marital_married	0.040732	...	-0.006270	0.005596	
marital_single	-0.059497	...	0.014372	-0.001389	
marital_unknown	0.003276	...	-0.005490	-0.004562	
education_basic.4y	0.184161	...	-0.004581	-0.011612	
education_basic.6y	0.009789	...	-0.014881	-0.004313	
education_basic.9y	-0.025442	...	-0.035151	0.007067	
education_high.school	-0.026086	...	-0.012765	0.007746	
education_illiterate	0.003928	...	-0.002618	0.001675	
education_professional.course	-0.035159	...	0.011927	-0.005288	
education_university.degree	-0.057079	...	0.028652	-0.006144	
education_unknown	-0.001847	...	0.028056	0.015976	

default_no	-0.036834	...	0.056497	-0.003246
default_unknown	0.036870	...	-0.056481	0.003338
default_yes	-0.001447	...	-0.001069	-0.004283
housing_no	0.002975	...	-0.007098	0.013482
housing_unknown	0.003277	...	0.002415	0.005591
housing_yes	-0.003994	...	0.006315	-0.015191
loan_no	0.002919	...	0.003204	-0.007280
loan_unknown	0.003277	...	0.002415	0.005591
loan_yes	-0.004502	...	-0.004434	0.005319
contact_cellular	-0.010804	...	0.054370	-0.040438
contact_telephone	0.010804	...	-0.054370	0.040438
month_apr	-0.012777	...	-0.032075	0.026532
month_aug	0.022693	...	-0.049545	-0.019750
month_dec	0.012350	...	-0.008321	-0.010273
month_jul	0.020675	...	-0.055162	-0.057240
month_jun	-0.001455	...	-0.047616	0.024838
month_mar	-0.010999	...	-0.014276	-0.005193
month_may	-0.018056	...	-0.086542	0.032052
month_nov	-0.008994	...	-0.040632	-0.003788
month_oct	0.000469	...	-0.016547	0.002795
month_sep	-0.003855	...	1.000000	0.003238
day_of_week_fri	-0.007712	...	0.003238	1.000000
day_of_week_mon	0.002484	...	-0.015863	-0.250373
day_of_week_thu	-0.009665	...	0.002023	-0.249358
day_of_week_tue	0.013182	...	0.003739	-0.239829
day_of_week_wed	0.001816	...	0.007237	-0.240557
poutcome_failure	-0.018584	...	0.052494	0.018460
poutcome_nonexistent	0.015602	...	-0.125303	-0.008483
poutcome_success	0.001598	...	0.149293	-0.014871
Loan_Status_label	-0.007496	...	0.125469	-0.008518

	day_of_week_mon	day_of_week_thu \
age	0.017340	-0.019094
campaign	0.010532	0.007932
pdays	0.002517	-0.009324
previous	-0.004078	0.002367
no_previous_contact	0.002493	-0.009325
not_working	-0.006026	-0.001012
job_admin.	-0.000429	-0.003825
job_blue-collar	-0.009880	-0.007824
job_entrepreneur	0.005297	0.004588
job_housemaid	0.002484	-0.009665
job_management	0.008108	-0.005448
job_retired	-0.001843	-0.012078
job_self-employed	0.003905	0.012441
job_services	0.003081	0.001805
job_student	-0.002748	0.005998

job_technician	0.001060	0.011664
job_unemployed	-0.006045	0.008179
job_unknown	0.005743	0.001870
marital_divorced	0.005642	-0.009133
marital_married	0.002925	-0.004681
marital_single	-0.007547	0.010514
marital_unknown	0.003487	0.010693
education_basic.4y	-0.003697	-0.001962
education_basic.6y	-0.013070	0.002004
education_basic.9y	-0.001487	-0.005331
education_high.school	0.007720	-0.016786
education_illiterate	-0.008138	0.006669
education_professional.course	-0.004363	0.010447
education_university.degree	0.008313	0.011993
education_unknown	-0.003658	0.000158
default_no	-0.007950	0.006681
default_unknown	0.008048	-0.006585
default_yes	-0.004523	-0.004505
housing_no	-0.009761	-0.008693
housing_unknown	-0.006765	-0.006249
housing_yes	0.011852	0.010625
loan_no	-0.001360	0.003302
loan_unknown	-0.006765	-0.006249
loan_yes	0.004346	-0.000818
contact_cellular	0.017729	0.037843
contact_telephone	-0.017729	-0.037843
month_apr	0.037969	0.052326
month_aug	-0.006720	0.007703
month_dec	0.014094	0.006893
month_jul	0.008144	0.023512
month_jun	0.026228	-0.027991
month_mar	0.015640	-0.007520
month_may	-0.027928	-0.038689
month_nov	-0.018047	0.008590
month_oct	-0.009317	0.006801
month_sep	-0.015863	0.002023
day_of_week_fri	-0.250373	-0.249358
day_of_week_mon	1.000000	-0.263350
day_of_week_thu	-0.263350	1.000000
day_of_week_tue	-0.253287	-0.252260
day_of_week_wed	-0.254056	-0.253026
poutcome_failure	-0.002989	-0.004108
poutcome_nonexistent	0.004625	-0.001620
poutcome_success	-0.003749	0.009951
Loan_Status_label	-0.023220	0.015907

day_of_week_tue day_of_week_wed \

age	0.017902	-0.022350
campaign	-0.024070	-0.020438
pdays	-0.008213	0.000339
previous	0.001799	-0.003039
no_previous_contact	-0.008194	0.000353
not_working	0.007139	-0.001549
job_admin.	-0.000034	-0.003369
job_blue-collar	-0.004312	0.019503
job_entrepreneur	-0.005552	-0.003921
job_housemaid	0.013182	0.001816
job_management	0.000223	0.002314
job_retired	0.009712	0.002480
job_self-employed	-0.007180	-0.013237
job_services	-0.001887	-0.004428
job_student	-0.000699	-0.000613
job_technician	-0.000141	-0.008075
job_unemployed	0.001115	-0.005452
job_unknown	0.006247	0.003874
marital_divorced	0.011422	-0.001909
marital_married	-0.000793	-0.002968
marital_single	-0.006604	0.004955
marital_unknown	-0.006385	-0.003601
education_basic.4y	0.016386	0.000943
education_basic.6y	0.003332	0.012286
education_basic.9y	-0.012025	0.011968
education_high.school	-0.001943	0.003506
education_illiterate	0.004487	-0.004618
education_professional.course	0.008870	-0.009819
education_university.degree	-0.009800	-0.004895
education_unknown	0.003702	-0.015945
default_no	-0.002180	0.006708
default_unknown	0.001795	-0.006615
default_yes	0.017859	-0.004346
housing_no	0.006668	-0.001141
housing_unknown	0.006574	0.001208
housing_yes	-0.008710	0.000757
loan_no	0.004822	0.000422
loan_unknown	0.006574	0.001208
loan_yes	-0.007935	-0.000966
contact_cellular	-0.004198	-0.012550
contact_telephone	0.004198	0.012550
month_apr	-0.064648	-0.054110
month_aug	0.012964	0.005645
month_dec	-0.010825	-0.000489
month_jul	0.017680	0.006722
month_jun	-0.010127	-0.012771
month_mar	0.018475	-0.021628

month_may	0.010878	0.025502
month_nov	0.003359	0.010107
month_oct	0.001535	-0.001710
month_sep	0.003739	0.007237
day_of_week_fri	-0.239829	-0.240557
day_of_week_mon	-0.253287	-0.254056
day_of_week_thu	-0.252260	-0.253026
day_of_week_tue	1.000000	-0.243357
day_of_week_wed	-0.243357	1.000000
poutcome_failure	-0.004376	-0.006670
poutcome_nonexistent	-0.000427	0.005753
poutcome_success	0.008142	0.000284
Loan_Status_label	0.009548	0.006439

	poutcome_failure	poutcome_nonexistent \
age	-0.007085	-0.011749
campaign	-0.075279	0.096142
pdays	0.009098	0.493971
previous	0.678092	-0.877385
no_previous_contact	0.008844	0.493975
not_working	0.031470	-0.082430
job_admin.	0.005870	-0.020046
job_blue-collar	-0.014362	0.045937
job_entrepreneur	-0.000734	0.010381
job_housemaid	-0.018584	0.015602
job_management	0.008590	-0.006468
job_retired	0.020400	-0.053786
job_self-employed	-0.001428	0.008759
job_services	0.010182	0.006644
job_student	0.040981	-0.080117
job_technician	-0.016188	0.016525
job_unemployed	-0.007025	-0.006388
job_unknown	-0.016281	0.008305
marital_divorced	0.005153	0.001920
marital_married	-0.033348	0.045662
marital_single	0.031867	-0.050052
marital_unknown	0.008346	-0.010990
education_basic.4y	-0.020362	0.021362
education_basic.6y	-0.006848	0.018106
education_basic.9y	-0.001092	0.019982
education_high.school	0.025623	-0.018357
education_illiterate	-0.003607	0.001950
education_professional.course	-0.010082	0.006952
education_university.degree	0.001373	-0.022744
education_unknown	0.000571	-0.009727
default_no	0.076248	-0.109128
default_unknown	-0.076397	0.109246

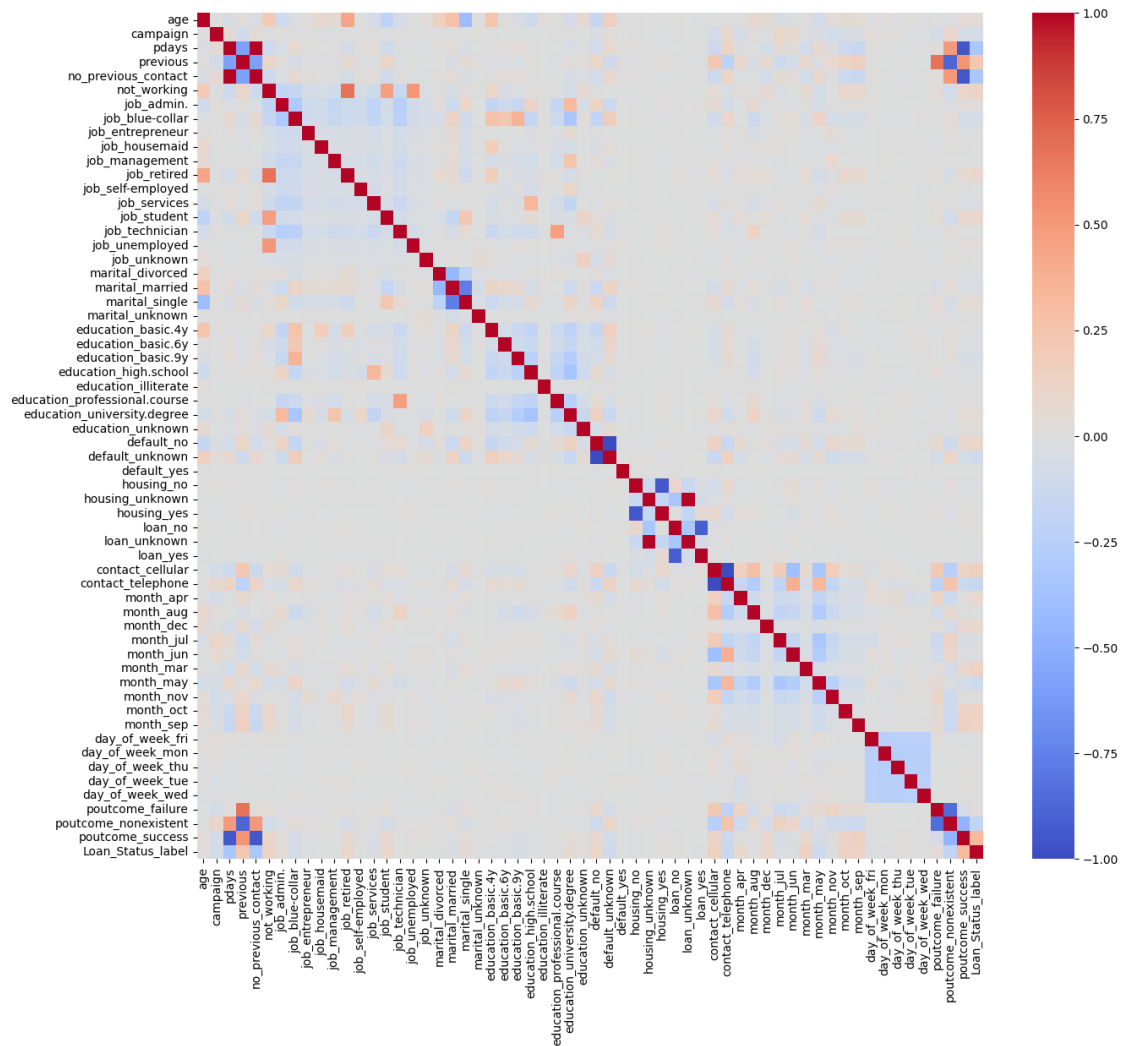
default_yes	0.006421	-0.004792
housing_no	-0.021451	0.025396
housing_unknown	0.002061	0.001299
housing_yes	0.020729	-0.025715
loan_no	0.003634	-0.006530
loan_unknown	0.002061	0.001299
loan_yes	-0.004738	0.006366
contact_cellular	0.215445	-0.252869
contact_telephone	-0.215445	0.252869
month_apr	0.121114	-0.113639
month_aug	-0.080910	0.069836
month_dec	0.023913	-0.062906
month_jul	-0.130906	0.140620
month_jun	-0.093989	0.091570
month_mar	0.029692	-0.065918
month_may	0.057435	-0.014661
month_nov	0.111967	-0.105703
month_oct	0.055576	-0.110399
month_sep	0.052494	-0.125303
day_of_week_fri	0.018460	-0.008483
day_of_week_mon	-0.002989	0.004625
day_of_week_thu	-0.004108	-0.001620
day_of_week_tue	-0.004376	-0.000427
day_of_week_wed	-0.006670	0.005753
poutcome_failure	1.000000	-0.849886
poutcome_nonexistent	-0.849886	1.000000
poutcome_success	-0.066430	-0.469345
Loan_Status_label	0.028345	-0.191993

	poutcome_success	Loan_Status_label
age	0.034118	0.026865
campaign	-0.055900	-0.074539
pdays	-0.950559	-0.324611
previous	0.525045	0.228665
no_previous_contact	-0.950141	-0.324572
not_working	0.103344	0.118294
job_admin.	0.028119	0.035250
job_blue-collar	-0.062914	-0.075625
job_entrepreneur	-0.018425	-0.019306
job_housemaid	0.001598	-0.007496
job_management	-0.002147	-0.001627
job_retired	0.067659	0.091108
job_self-employed	-0.014192	-0.006224
job_services	-0.029641	-0.033042
job_student	0.083029	0.092536
job_technician	-0.004164	-0.003678
job_unemployed	0.023866	0.012016

job_unknown	0.011555	-0.002082
marital_divorced	-0.012270	-0.013390
marital_married	-0.030579	-0.043307
marital_single	0.041373	0.056264
marital_unknown	0.006824	0.004679
education_basic.4y	-0.006328	-0.011243
education_basic.6y	-0.022808	-0.024966
education_basic.9y	-0.036006	-0.047056
education_high.school	-0.008176	-0.008653
education_illiterate	0.002353	0.006922
education_professional.course	0.003730	0.002225
education_university.degree	0.040765	0.054144
education_unknown	0.017460	0.019507
default_no	0.078865	0.103282
default_unknown	-0.078839	-0.103227
default_yes	-0.001685	-0.003224
housing_no	-0.012142	-0.011295
housing_unknown	-0.005912	-0.004555
housing_yes	0.013956	0.012686
loan_no	0.006275	0.011501
loan_unknown	-0.005912	-0.004555
loan_yes	-0.004116	-0.010239
contact_cellular	0.117786	0.152383
contact_telephone	-0.117786	-0.152383
month_apr	0.012225	0.075303
month_aug	0.003346	-0.001080
month_dec	0.079041	0.077943
month_jul	-0.046906	-0.030064
month_jun	-0.015890	-0.013482
month_mar	0.075059	0.141504
month_may	-0.068481	-0.112099
month_nov	0.012527	-0.014042
month_oct	0.115910	0.135743
month_sep	0.149293	0.125469
day_of_week_fri	-0.014871	-0.008518
day_of_week_mon	-0.003749	-0.023220
day_of_week_thu	0.009951	0.015907
day_of_week_tue	0.008142	0.009548
day_of_week_wed	0.000284	0.006439
poutcome_failure	-0.066430	0.028345
poutcome_nonexistent	-0.469345	-0.191993
poutcome_success	1.000000	0.316035
Loan_Status_label	0.316035	1.000000

[60 rows x 60 columns]

```
[22]: plt.figure(figsize=(15,13))
sns.heatmap(df.corr(),cmap='coolwarm')
plt.show()
```



Model Building

```
[23]: #Separating independent and dependent data

x=df.drop(columns=["Loan_Status_label"], axis=1)
y=df["Loan_Status_label"]
```

```
[24]: x
```

```
[24]:    age  campaign  pdays  previous  no_previous_contact  not_working  \
0     56         1    999         0                   1             0
```


1	57	1	999	0	1	0
2	37	1	999	0	1	0
3	40	1	999	0	1	0
4	56	1	999	0	1	0
...	...	...	...	...	...	...
41183	73	1	999	0	1	1
41184	46	1	999	0	1	0
41185	56	2	999	0	1	1
41186	44	1	999	0	1	0
41187	74	3	999	1	1	1

	job_admin.	job_blue-collar	job_entrepreneur	job_housemaid	...	\
0	0	0	0	1	...	
1	0	0	0	0	...	
2	0	0	0	0	...	
3	1	0	0	0	...	
4	0	0	0	0	...	
...	...	...	...	...	...	
41183	0	0	0	0	...	
41184	0	1	0	0	...	
41185	0	0	0	0	...	
41186	0	0	0	0	...	
41187	0	0	0	0	...	

	month_oct	month_sep	day_of_week_fri	day_of_week_mon	\
0	0	0	0	1	
1	0	0	0	1	
2	0	0	0	1	
3	0	0	0	1	
4	0	0	0	1	
...	...	...	...	...	
41183	0	0	1	0	
41184	0	0	1	0	
41185	0	0	1	0	
41186	0	0	1	0	
41187	0	0	1	0	

	day_of_week_thu	day_of_week_tue	day_of_week_wed	poutcome_failure	\
0	0	0	0	0	
1	0	0	0	0	
2	0	0	0	0	
3	0	0	0	0	
4	0	0	0	0	
...	...	...	...	...	
41183	0	0	0	0	
41184	0	0	0	0	
41185	0	0	0	0	

41186	0	0	0	0
41187	0	0	0	1

	poutcome_nonexistent	poutcome_success
0	1	0
1	1	0
2	1	0
3	1	0
4	1	0
...	...	...
41183	1	0
41184	1	0
41185	1	0
41186	1	0
41187	0	0

[38771 rows x 59 columns]

[25]: y

```
[25]: 0      0
      1      0
      2      0
      3      0
      4      0
      ..
      41183  1
      41184  0
      41185  0
      41186  1
      41187  0
```

Name: Loan_Status_label, Length: 38771, dtype: int64

[26]: *#Splitting data into train & test*

```
X_train, X_test, y_train, y_test = train_test_split(x, y, test_size=0.2,
↳ random_state=42)
```

[27]: X_train

```
[27]:   age  campaign  pdays  previous  no_previous_contact  not_working  \
34767   38         9    999         0                   1           0
32132   36         1    999         1                   1           0
9868    49         1    999         0                   1           0
20192   36         9    999         0                   1           0
28805   50         2    999         1                   1           0
...    ...      ...    ...      ...                   ...           ...
```

6625	35	2	999	0	1	1
11843	31	3	999	0	1	0
40568	70	4	999	0	1	1
885	43	1	999	0	1	0
16784	43	2	999	0	1	0

	job_admin.	job_blue-collar	job_entrepreneur	job_housemaid	...	\
34767	0	1	0	0	...	
32132	0	0	0	0	...	
9868	0	1	0	0	...	
20192	0	0	0	1	...	
28805	0	1	0	0	...	
...	...	...	...	...	...	
6625	0	0	0	0	...	
11843	1	0	0	0	...	
40568	0	0	0	0	...	
885	0	0	0	0	...	
16784	0	0	0	1	...	

	month_oct	month_sep	day_of_week_fri	day_of_week_mon	\
34767	0	0	0	0	
32132	0	0	1	0	
9868	0	0	0	0	
20192	0	0	0	1	
28805	0	0	0	0	
...	...	...	...	...	
6625	0	0	0	0	
11843	0	0	1	0	
40568	0	1	0	0	
885	0	0	0	0	
16784	0	0	0	0	

	day_of_week_thu	day_of_week_tue	day_of_week_wed	poutcome_failure	\
34767	1	0	0	0	
32132	0	0	0	1	
9868	0	0	1	0	
20192	0	0	0	0	
28805	1	0	0	1	
...	...	...	...	...	
6625	0	0	1	0	
11843	0	0	0	0	
40568	0	0	1	0	
885	0	0	1	0	
16784	1	0	0	0	

	poutcome_nonexistent	poutcome_success
34767	1	0

32132	0	0
9868	1	0
20192	1	0
28805	0	0
...	...	...
6625	1	0
11843	1	0
40568	1	0
885	1	0
16784	1	0

[31016 rows x 59 columns]

[28]: X_test

```
[28]:
```

	age	campaign	pdays	previous	no_previous_contact	not_working	\
37815	57	1	999	1	1	1	
5279	53	2	999	0	1	1	
40810	30	1	999	3	1	0	
3788	29	2	999	0	1	0	
20910	30	1	999	0	1	0	
...	...	...	...	...	...	...	
6833	35	2	999	0	1	0	
18429	45	1	999	0	1	0	
18590	28	2	999	0	1	0	
4545	33	10	999	0	1	0	
26987	33	3	999	0	1	0	

	job_admin.	job_blue-collar	job_entrepreneur	job_housemaid	...	\
37815	0	0	0	0	...	
5279	0	0	0	0	...	
40810	0	0	0	0	...	
3788	0	0	0	0	...	
20910	0	0	0	0	...	
...	...	...	...	...	...	
6833	0	1	0	0	...	
18429	0	1	0	0	...	
18590	0	0	0	0	...	
4545	0	1	0	0	...	
26987	1	0	0	0	...	

	month_oct	month_sep	day_of_week_fri	day_of_week_mon	\
37815	0	0	0	0	
5279	0	0	1	0	
40810	0	1	0	0	
3788	0	0	1	0	
20910	0	0	0	0	

...	...	...	...	...	...
6833	0	0	0	0	0
18429	0	0	0	0	0
18590	0	0	0	0	0
4545	0	0	0	0	0
26987	0	0	0	0	0

	day_of_week_thu	day_of_week_tue	day_of_week_wed	poutcome_failure	\
37815	0	0	1	1	
5279	0	0	0	0	
40810	1	0	0	1	
3788	0	0	0	0	
20910	1	0	0	0	

...	...	...	...	...	
6833	0	0	1	0	
18429	1	0	0	0	
18590	1	0	0	0	
4545	0	1	0	0	
26987	1	0	0	0	

	poutcome_nonexistent	poutcome_success
37815	0	0
5279	1	0
40810	0	0
3788	1	0
20910	1	0

...	...	...
6833	1	0
18429	1	0
18590	1	0
4545	1	0
26987	1	0

[7755 rows x 59 columns]

Feature Scaling

```
[29]: from sklearn.preprocessing import StandardScaler
```

```
[30]: sc = StandardScaler()
X_train_sc = sc.fit_transform(X_train)
X_test_sc = sc.transform(X_test)
```

```
[31]: X_train
```

```
[31]:
```

	age	campaign	pdays	previous	no_previous_contact	not_working	\
34767	38	9	999	0	1	0	
32132	36	1	999	1	1	0	

9868	49	1	999	0	1	0
20192	36	9	999	0	1	0
28805	50	2	999	1	1	0
...	...	...	...	...	...	...
6625	35	2	999	0	1	1
11843	31	3	999	0	1	0
40568	70	4	999	0	1	1
885	43	1	999	0	1	0
16784	43	2	999	0	1	0

	job_admin.	job_blue-collar	job_entrepreneur	job_housemaid	...	\
34767	0	1	0	0	...	
32132	0	0	0	0	...	
9868	0	1	0	0	...	
20192	0	0	0	1	...	
28805	0	1	0	0	...	
...	...	...	...	...	...	
6625	0	0	0	0	...	
11843	1	0	0	0	...	
40568	0	0	0	0	...	
885	0	0	0	0	...	
16784	0	0	0	1	...	

	month_oct	month_sep	day_of_week_fri	day_of_week_mon	\
34767	0	0	0	0	
32132	0	0	1	0	
9868	0	0	0	0	
20192	0	0	0	1	
28805	0	0	0	0	
...	...	...	...	...	
6625	0	0	0	0	
11843	0	0	1	0	
40568	0	1	0	0	
885	0	0	0	0	
16784	0	0	0	0	

	day_of_week_thu	day_of_week_tue	day_of_week_wed	poutcome_failure	\
34767	1	0	0	0	
32132	0	0	0	1	
9868	0	0	1	0	
20192	0	0	0	0	
28805	1	0	0	1	
...	...	...	...	...	
6625	0	0	1	0	
11843	0	0	0	0	
40568	0	0	1	0	
885	0	0	1	0	

16784	1	0	0	0
-------	---	---	---	---

	poutcome_nonexistent	poutcome_success
34767	1	0
32132	0	0
9868	1	0
20192	1	0
28805	0	0
...	...	...
6625	1	0
11843	1	0
40568	1	0
885	1	0
16784	1	0

[31016 rows x 59 columns]

```
[32]: X_train_sc
```

```
[32]: array([[ -0.21204265,  2.27564026,  0.20500869, ..., -0.34651856,
          0.40992066, -0.19507302],
        [ -0.40249266, -0.58081638,  0.20500869, ...,  2.88584828,
        -2.43949646, -0.19507302],
        [  0.8354324 , -0.58081638,  0.20500869, ..., -0.34651856,
          0.40992066, -0.19507302],
        ...,
        [  2.83515749,  0.49035486,  0.20500869, ..., -0.34651856,
          0.40992066, -0.19507302],
        [  0.26408237, -0.58081638,  0.20500869, ..., -0.34651856,
          0.40992066, -0.19507302],
        [  0.26408237, -0.2237593 ,  0.20500869, ..., -0.34651856,
          0.40992066, -0.19507302]], shape=(31016, 59))
```

```
[33]: X_test_sc
```

```
[33]: array([[ 1.59723243, -0.58081638,  0.20500869, ...,  2.88584828,
        -2.43949646, -0.19507302],
        [ 1.21633241, -0.2237593 ,  0.20500869, ..., -0.34651856,
          0.40992066, -0.19507302],
        [-0.97384269, -0.58081638,  0.20500869, ...,  2.88584828,
        -2.43949646, -0.19507302],
        ...,
        [-1.1642927 , -0.2237593 ,  0.20500869, ..., -0.34651856,
          0.40992066, -0.19507302],
        [-0.68816768,  2.63269734,  0.20500869, ..., -0.34651856,
          0.40992066, -0.19507302],
        [-0.68816768,  0.13329778,  0.20500869, ..., -0.34651856,
```

```
0.40992066, -0.19507302]], shape=(7755, 59))
```

Model Selection

```
[39]: #Logistic Regression

lg = LogisticRegression(max_iter=1000)
lg.fit(X_train_sc, y_train)
y_pred_log = lg.predict(X_test_sc)
print(f'Training Accuracy : {round(lg.score(X_train_sc,y_train),2)*100}%')
print(f'Test Accuracy : {round(lg.score(X_test_sc,y_test),2)*100}%')
```

Training Accuracy : 89.0%

Test Accuracy : 90.0%

```
[44]: #Decision Tree

dtree=DecisionTreeClassifier(max_depth=6)
dtree.fit(X_train_sc,y_train)
print(f'Training Accuracy : {round(dtree.score(X_train_sc,y_train),2)*100}%')
print(f'Test Accuracy : {round(dtree.score(X_test_sc,y_test),2)*100}%')
```

Training Accuracy : 89.0%

Test Accuracy : 90.0%

```
[45]: dtree=DecisionTreeClassifier(max_depth=5)
dtree.fit(X_train_sc,y_train)
```

```
[45]: DecisionTreeClassifier(max_depth=5)
```

Model Prediction

```
[50]: y_pred_train = dtree.predict(X_train_sc)
y_pred_test = dtree.predict(X_test_sc)
X_train[:3]
```

```
[50]:
```

	age	campaign	pdays	previous	no_previous_contact	not_working	\
34767	38	9	999	0	1	0	
32132	36	1	999	1	1	0	
9868	49	1	999	0	1	0	

	job_admin.	job_blue-collar	job_entrepreneur	job_housemaid	...	\
34767	0	1	0	0	...	
32132	0	0	0	0	...	
9868	0	1	0	0	...	

	month_oct	month_sep	day_of_week_fri	day_of_week_mon	\
34767	0	0	0	0	
32132	0	0	1	0	

9868	0	0	0	0
------	---	---	---	---

	day_of_week_thu	day_of_week_tue	day_of_week_wed	poutcome_failure \
34767	1	0	0	0
32132	0	0	0	1
9868	0	0	1	0

	poutcome_nonexistent	poutcome_success
34767	1	0
32132	0	0
9868	1	0

[3 rows x 59 columns]

```
[49]: y_train[:3]
```

```
[49]: 34767    0
      32132    0
      9868    0
      Name: Loan_Status_label, dtype: int64
```

```
[51]: y_pred_train[:3]
```

```
[51]: array([0, 0, 0])
```

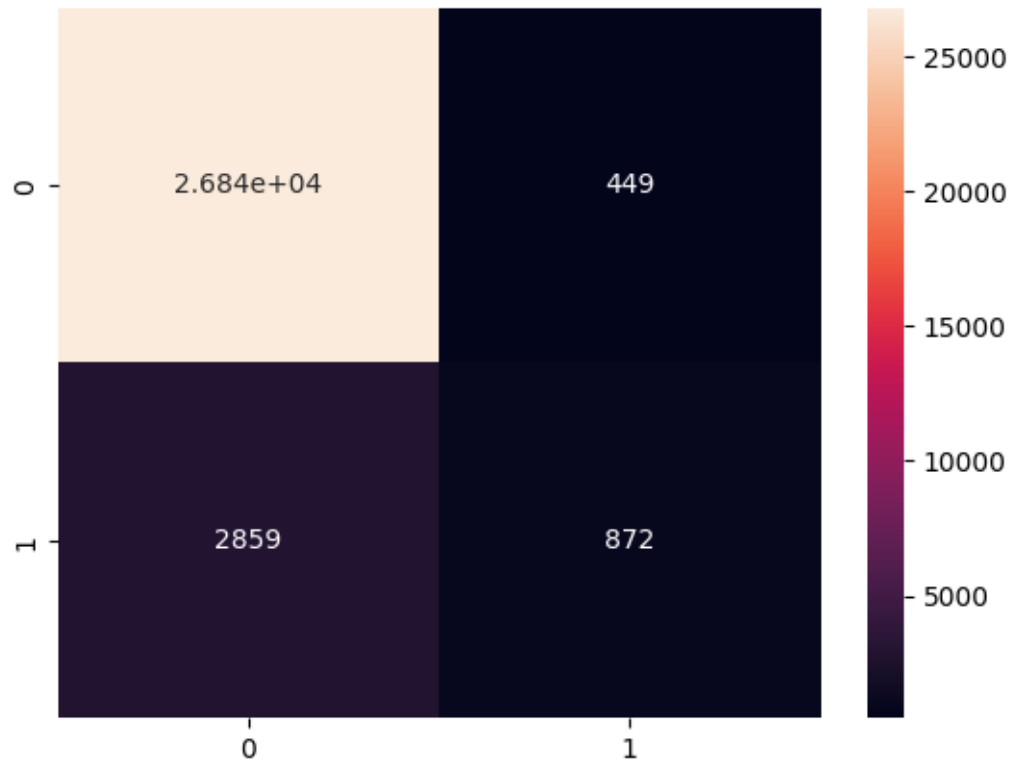
Model Evaluation

```
[53]: confusion_matrix(y_train,y_pred_train)
```

```
[53]: array([[26836,  449],
          [ 2859,  872]])
```

```
[55]: sns.heatmap(confusion_matrix(y_train,y_pred_train),annot=True,fmt='.4g')
```

```
[55]: <Axes: >
```



```
[57]: accuracy_score(y_train,y_pred_train)
```

```
[57]: 0.893345370131545
```

```
[63]: print(classification_report(y_train, y_pred_train))
```

	precision	recall	f1-score	support
0	0.90	0.98	0.94	27285
1	0.66	0.23	0.35	3731
accuracy			0.89	31016
macro avg	0.78	0.61	0.64	31016
weighted avg	0.87	0.89	0.87	31016

```
[ ]:
```

SAMPLE TEST

```
[64]: x[:1]
```

```
[64]: age campaign pdays previous no_previous_contact not_working \
0 56 1 999 0 1 0

job_admin. job_blue-collar job_entrepreneur job_housemaid ... \
0 0 0 0 1 ...

month_oct month_sep day_of_week_fri day_of_week_mon day_of_week_thu \
0 0 0 0 1 0

day_of_week_tue day_of_week_wed poutcome_failure poutcome_nonexistent \
0 0 0 0 1

poutcome_success
0 0

[1 rows x 59 columns]
```

```
[65]: y[:1]
```

```
[65]: 0 0
Name: Loan_Status_label, dtype: int64
```

```
[66]: x.ndim
```

```
[66]: 2
```

```
[69]: new_data =
↳ [[29,1,999,0,1,0,0,0,0,0,0,0,0,0,0,0,0,1,0,0,0,0,1,0,0,0,0,1,0,0,1,0,0,0,0,1,0,1,0,0,0,
new_data
```

```
[69]: [[29,
1,
999,
0,
1,
0,
0,
0,
0,
0,
0,
0,
0,
0,
0,
0,
0,
1,
0,
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1,
0,
0,
0,
```

```
0,  
0,  
1,  
0,  
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1,  
0,  
0,  
0,  
0,  
1,  
0,  
0,  
0,  
1,  
0,  
0]]
```

```
[70]: dtree.predict(new_data)
```

```
[70]: array([0])
```


nbconvert->notebook-as-pdf) (5.10.4)
 Requirement already satisfied: packaging in
 c:\users\pari\appdata\local\programs\python\python313\lib\site-packages (from
 nbconvert->notebook-as-pdf) (24.2)
 Requirement already satisfied: pandocfilters>=1.4.1 in
 c:\users\pari\appdata\local\programs\python\python313\lib\site-packages (from
 nbconvert->notebook-as-pdf) (1.5.1)
 Requirement already satisfied: pygments>=2.4.1 in
 c:\users\pari\appdata\local\programs\python\python313\lib\site-packages (from
 nbconvert->notebook-as-pdf) (2.19.1)
 Requirement already satisfied: traitlets>=5.1 in
 c:\users\pari\appdata\local\programs\python\python313\lib\site-packages (from
 nbconvert->notebook-as-pdf) (5.14.3)
 Requirement already satisfied: webencodings in
 c:\users\pari\appdata\local\programs\python\python313\lib\site-packages (from
 bleach!=5.0.0->bleach[css]!=5.0.0->nbconvert->notebook-as-pdf) (0.5.1)
 Requirement already satisfied: tinycss2<1.5,>=1.1.0 in
 c:\users\pari\appdata\local\programs\python\python313\lib\site-packages (from
 bleach[css]!=5.0.0->nbconvert->notebook-as-pdf) (1.4.0)
 Requirement already satisfied: platformdirs>=2.5 in
 c:\users\pari\appdata\local\programs\python\python313\lib\site-packages (from
 jupyter-core>=4.7->nbconvert->notebook-as-pdf) (4.3.7)
 Requirement already satisfied: pywin32>=300 in
 c:\users\pari\appdata\local\programs\python\python313\lib\site-packages (from
 jupyter-core>=4.7->nbconvert->notebook-as-pdf) (310)
 Requirement already satisfied: jupyter-client>=6.1.12 in
 c:\users\pari\appdata\local\programs\python\python313\lib\site-packages (from
 nbclient>=0.5.0->nbconvert->notebook-as-pdf) (8.6.3)
 Requirement already satisfied: python-dateutil>=2.8.2 in
 c:\users\pari\appdata\local\programs\python\python313\lib\site-packages (from
 jupyter-client>=6.1.12->nbclient>=0.5.0->nbconvert->notebook-as-pdf)
 (2.9.0.post0)
 Requirement already satisfied: pyzmq>=23.0 in
 c:\users\pari\appdata\local\programs\python\python313\lib\site-packages (from
 jupyter-client>=6.1.12->nbclient>=0.5.0->nbconvert->notebook-as-pdf) (26.4.0)
 Requirement already satisfied: tornado>=6.2 in
 c:\users\pari\appdata\local\programs\python\python313\lib\site-packages (from
 jupyter-client>=6.1.12->nbclient>=0.5.0->nbconvert->notebook-as-pdf) (6.4.2)
 Requirement already satisfied: fastjsonschema>=2.15 in
 c:\users\pari\appdata\local\programs\python\python313\lib\site-packages (from
 nbformat>=5.7->nbconvert->notebook-as-pdf) (2.21.1)
 Requirement already satisfied: jsonschema>=2.6 in
 c:\users\pari\appdata\local\programs\python\python313\lib\site-packages (from
 nbformat>=5.7->nbconvert->notebook-as-pdf) (4.23.0)
 Requirement already satisfied: attrs>=22.2.0 in
 c:\users\pari\appdata\local\programs\python\python313\lib\site-packages (from
 jsonschema>=2.6->nbformat>=5.7->nbconvert->notebook-as-pdf) (25.3.0)
 Requirement already satisfied: jsonschema-specifications>=2023.03.6 in

```

c:\users\pari\appdata\local\programs\python\python313\lib\site-packages (from
jsonschema>=2.6->nbformat>=5.7->nbconvert->notebook-as-pdf) (2024.10.1)
Requirement already satisfied: referencing>=0.28.4 in
c:\users\pari\appdata\local\programs\python\python313\lib\site-packages (from
jsonschema>=2.6->nbformat>=5.7->nbconvert->notebook-as-pdf) (0.36.2)
Requirement already satisfied: rpds-py>=0.7.1 in
c:\users\pari\appdata\local\programs\python\python313\lib\site-packages (from
jsonschema>=2.6->nbformat>=5.7->nbconvert->notebook-as-pdf) (0.24.0)
Requirement already satisfied: six>=1.5 in
c:\users\pari\appdata\local\programs\python\python313\lib\site-packages (from
python-dateutil>=2.8.2->jupyter-
client>=6.1.12->nbclient>=0.5.0->nbconvert->notebook-as-pdf) (1.17.0)
Requirement already satisfied: soupsieve>1.2 in
c:\users\pari\appdata\local\programs\python\python313\lib\site-packages (from
beautifulsoup4->nbconvert->notebook-as-pdf) (2.6)
Requirement already satisfied: typing-extensions>=4.0.0 in
c:\users\pari\appdata\local\programs\python\python313\lib\site-packages (from
beautifulsoup4->nbconvert->notebook-as-pdf) (4.13.1)
Collecting appdirs<2.0.0,>=1.4.3 (from pyppeteer->notebook-as-pdf)
  Downloading appdirs-1.4.4-py2.py3-none-any.whl.metadata (9.0 kB)
Requirement already satisfied: certifi>=2023 in
c:\users\pari\appdata\local\programs\python\python313\lib\site-packages (from
pyppeteer->notebook-as-pdf) (2025.1.31)
Collecting importlib-metadata>=1.4 (from pyppeteer->notebook-as-pdf)
  Downloading importlib_metadata-8.7.0-py3-none-any.whl.metadata (4.8 kB)
Collecting pyee<12.0.0,>=11.0.0 (from pyppeteer->notebook-as-pdf)
  Downloading pyee-11.1.1-py3-none-any.whl.metadata (2.8 kB)
Requirement already satisfied: tqdm<5.0.0,>=4.42.1 in
c:\users\pari\appdata\local\programs\python\python313\lib\site-packages (from
pyppeteer->notebook-as-pdf) (4.67.1)
Collecting urllib3<2.0.0,>=1.25.8 (from pyppeteer->notebook-as-pdf)
  Downloading urllib3-1.26.20-py2.py3-none-any.whl.metadata (50 kB)
Collecting websockets<11.0,>=10.0 (from pyppeteer->notebook-as-pdf)
  Downloading websockets-10.4.tar.gz (84 kB)
Installing build dependencies: started
Installing build dependencies: finished with status 'done'
Getting requirements to build wheel: started
Getting requirements to build wheel: finished with status 'done'
Preparing metadata (pyproject.toml): started
Preparing metadata (pyproject.toml): finished with status 'done'
Requirement already satisfied: colorama in
c:\users\pari\appdata\local\programs\python\python313\lib\site-packages (from
tqdm<5.0.0,>=4.42.1->pyppeteer->notebook-as-pdf) (0.4.6)
Collecting zipp>=3.20 (from importlib-metadata>=1.4->pyppeteer->notebook-as-pdf)
  Downloading zipp-3.23.0-py3-none-any.whl.metadata (3.6 kB)
Downloading notebook_as_pdf-0.5.0-py3-none-any.whl (6.5 kB)
Downloading pypdf2-3.0.1-py3-none-any.whl (232 kB)
Downloading pyppeteer-2.0.0-py3-none-any.whl (82 kB)

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Downloading appdirs-1.4.4-py2.py3-none-any.whl (9.6 kB)
Downloading pyee-11.1.1-py3-none-any.whl (15 kB)
Downloading urllib3-1.26.20-py2.py3-none-any.whl (144 kB)
Downloading importlib_metadata-8.7.0-py3-none-any.whl (27 kB)
Downloading zipp-3.23.0-py3-none-any.whl (10 kB)
Building wheels for collected packages: websockets
  Building wheel for websockets (pyproject.toml): started
  Building wheel for websockets (pyproject.toml): finished with status 'done'
  Created wheel for websockets:
    filename=websockets-10.4-cp313-cp313-win_amd64.whl size=95113
    sha256=1e134ebbd67d10906bba99a7490b42e8b43bee8db571ab07ee5dd1fd5ee9e590
    Stored in directory: c:\users\pari\appdata\local\pip\cache\wheels\f5\dd\43\5b7
    132085406a982d2e0dd808f0ffed137865c8a8c5c7a5e4b
Successfully built websockets
Installing collected packages: appdirs, zipp, websockets, urllib3, PyPDF2, pyee,
importlib-metadata, pypeteer, notebook-as-pdf

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----- 1/9 [zipp]
----- 2/9 [websockets]
----- 2/9 [websockets]
----- 2/9 [websockets]
----- 2/9 [websockets]
----- 2/9 [websockets]
----- 2/9 [websockets]
Attempting uninstall: urllib3
----- 2/9 [websockets]
  Found existing installation: urllib3 2.3.0
----- 2/9 [websockets]
  Uninstalling urllib3-2.3.0:
----- 2/9 [websockets]
    Successfully uninstalled urllib3-2.3.0
----- 2/9 [websockets]
----- 3/9 [urllib3]
----- 3/9 [urllib3]
----- 3/9 [urllib3]
----- 3/9 [urllib3]
----- 3/9 [urllib3]
----- 3/9 [urllib3]
----- 3/9 [urllib3]
----- 3/9 [urllib3]
----- 3/9 [urllib3]
----- 3/9 [urllib3]
----- 3/9 [urllib3]
----- 4/9 [PyPDF2]
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----- 4/9 [PyPDF2]
----- 4/9 [PyPDF2]
----- 4/9 [PyPDF2]

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----- 4/9 [PyPDF2]
----- 5/9 [pyee]
----- 5/9 [pyee]
----- 6/9 [importlib-metadata]
----- 6/9 [importlib-metadata]
----- 7/9 [pyppeteer]
----- 7/9 [pyppeteer]
----- 7/9 [pyppeteer]
----- 7/9 [pyppeteer]
----- 7/9 [pyppeteer]
----- 7/9 [pyppeteer]
----- 7/9 [pyppeteer]
----- 7/9 [pyppeteer]
----- 9/9 [notebook-as-pdf]

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Successfully installed PyPDF2-3.0.1 appdirs-1.4.4 importlib-metadata-8.7.0
notebook-as-pdf-0.5.0 pyee-11.1.1 pyppeteer-2.0.0 urllib3-1.26.20
websockets-10.4 zipp-3.23.0

Note: you may need to restart the kernel to use updated packages.

[]:

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